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Guidelines

Diagnosis and Treatment of Non-Muscle Invasive Bladder Cancer: AUA/SUO Guideline

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(Published 2016; Amended 2020, 2024)

To cite this guideline:

Holzbeierlein J, Bixler BR, Buckley DI, et al. Diagnosis and treatment of non-muscle invasive bladder cancer: AUA/SUO guideline: 2024 amendment. J Urol. 2024;10.1097/JU.0000000000003846.
<https://www.auajournals.org/doi/10.1097/JU.0000000000003846>

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[AUA Risk Stratification for Non-Muscle Invasive Bladder Cancer](#)

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SUMMARY

Purpose

The survival rate for the majority of patients with non-muscle invasive bladder cancer (NMIBC) is favorable; however, the rates of recurrence and progression to muscle-invasive bladder cancer (MIBC) are important surrogate endpoints for overall prognosis, as these are major determinants of long-term outcome. The recurrence and progression probability rates depend on several clinical and pathologic factors. Therefore, the ability to predict risk of recurrence and progression and treat the disease appropriately is important. This guideline provides a risk-stratified clinical framework for the management of NMIBC. Please also refer to the associated NMIBC Treatment Algorithm.

Methodology

The systematic review utilized in the creation of this guideline was completed in part through the Agency for Healthcare Research and Quality (AHRQ) and through additional supplementation that further addressed additional key questions and more recently published literature. A research librarian experienced in conducting literature searches for comparative effectiveness reviews searched in Ovid MEDLINE (January 1990 – October 2014), Cochrane Central Register of Controlled Trials (through September 2014), Cochrane Database of Systematic Reviews (through September 2014), Health Technology Assessment (through 3rd Quarter, 2014), National Health Sciences Economic Evaluation Database (through 3rd Quarter, 2014), and Database of Abstracts of Reviews of Effects (through 3rd Quarter, 2014) to capture both published and grey literature. Database searches resulted in 3,740 potentially relevant articles. After dual review of abstracts and titles, 643 articles were selected for full-text dual review, and 149 studies (in 192 publications) were determined to meet inclusion criteria and were included in this review. The AHRQ review was then updated by a consultant methodologist though September 2, 2015. Reference lists and previous systematic reviews were also reviewed for additional studies. This supplementation added 29 studies to the completed systematic review used in the creation of guideline statements. The guideline underwent review in 2019. The updated search (June 1, 2015 to November 22, 2019) identified 1,626 abstracts, of which 76 met inclusion criteria. An additional review was performed in 2023. The updated search (July 2019 to May 2023) identified 1918 abstracts, of which 75 met inclusion criteria. When sufficient evidence existed, the body of evidence for a particular treatment was assigned a strength rating of A (high), B (moderate) or C (low) for support of Strong, Moderate, or Conditional Recommendations. In the absence of sufficient evidence, additional information is provided as Clinical Principles and Expert Opinions.

GUIDELINE STATEMENTS

Diagnosis

1. At the time of resection of suspected bladder cancer, a clinician should perform a thorough cystoscopic examination of a patient’s entire urethra and bladder that evaluates and documents tumor size, location, configuration, number, and mucosal abnormalities. (*Clinical Principle*)
2. At initial diagnosis of a patient with bladder cancer, a clinician should perform complete visual resection of the bladder tumor(s), when technically feasible. (*Clinical Principle*)
3. A clinician should perform upper urinary tract imaging as a component of the initial evaluation of a patient with bladder cancer. (*Clinical Principle*)
4. In a patient with a history of NMIBC with normal cystoscopy and positive cytology, a clinician should consider prostatic urethral biopsies and upper tract imaging, as well as enhanced cystoscopic techniques (blue light cystoscopy [BLC], when available), ureteroscopy, or random bladder biopsies. (*Expert Opinion*)

Risk Stratification

5. At the time of each occurrence/recurrence, a clinician should assign a clinical stage and classify a patient accordingly as “low-,” “intermediate-,” or “high-risk.” (*Moderate Recommendation; Evidence Strength: Grade C*)

Variant Histologies

6. An experienced genitourinary pathologist should review the pathology of a patient with any doubt in regard to variant or suspected variant histology (e.g., micropapillary, nested, plasmacytoid, neuroendocrine, sarcomatoid), extensive squamous or glandular differentiation, or the presence/absence of lymphovascular invasion (LVI). (*Moderate Recommendation; Evidence Strength: Grade C*)

7. If a bladder sparing approach is being considered in a patient with variant histology, then a clinician should perform a restaging transurethral resection of bladder tumor (TURBT) within four to six weeks of the initial TURBT. (*Expert Opinion*)
8. Due to the high rate of upstaging associated with variant histology, a clinician should consider offering initial radical cystectomy. (*Expert Opinion*)

Urine Markers After Diagnosis of Bladder Cancer

9. In surveillance of NMIBC, a clinician should not use urinary biomarkers in place of cystoscopic evaluation. (*Strong Recommendation; Evidence Strength: Grade B*)
10. In a patient with a history of low-risk cancer and a normal cystoscopy, a clinician should not routinely use a urinary biomarker or cytology during surveillance. (*Expert Opinion*)
11. In a patient with NMIBC, a clinician may use biomarkers to assess response to intravesical BCG (UroVysion® FISH) and adjudicate equivocal cytology (UroVysion® FISH and ImmunoCyt™). (*Expert Opinion*)

TURBT/Repeat Resection: Timing, Technique, Goal, Indication

12. In a patient with non-muscle invasive disease who underwent an incomplete initial resection (not all visible tumor treated), a clinician should perform repeat transurethral resection or endoscopic treatment of all remaining tumor if technically feasible. (*Strong Recommendation; Evidence Strength: Grade B*)
13. In a patient with high-risk, high-grade Ta tumors, a clinician should consider performing repeat transurethral resection of the primary tumor site within six weeks of the initial TURBT. (*Moderate Recommendation; Evidence Strength: Grade C*)
14. In a patient with T1 disease, a clinician should perform repeat transurethral resection of the primary tumor site to include muscularis propria within six weeks of the initial TURBT. (*Strong Recommendation; Evidence Strength: Grade B*)

Intravesical Therapy; BCG/Maintenance; Chemotherapy/BCG Combinations

15. In a patient with suspected or known low- or intermediate-risk bladder cancer, a clinician should consider administration of a single postoperative instillation of intravesical chemotherapy (e.g., gemcitabine, mitomycin C) within 24 hours of TURBT. In a patient with a suspected perforation or extensive resection, a clinician should not use postoperative intravesical chemotherapy. (*Moderate Recommendation; Evidence Strength: Grade B*)
16. In a low-risk patient, a clinician should not administer induction intravesical therapy. (*Moderate Recommendation; Evidence Strength: Grade C*)
17. In an intermediate-risk patient a clinician should consider administration of a six-week course of induction intravesical chemotherapy or immunotherapy. (*Moderate Recommendation; Evidence Strength: Grade B*)
18. In a high-risk patient with newly diagnosed carcinoma *in situ* (CIS), high-grade T1, or high-risk Ta urothelial carcinoma, a clinician should administer a six-week induction course of BCG. (*Strong Recommendation; Evidence Strength: Grade B*)
19. In an intermediate-risk patient who completely responds to an induction course of intravesical chemotherapy, a clinician may utilize maintenance therapy. (*Conditional Recommendation; Evidence Strength: Grade C*)
20. In an intermediate-risk patient who completely responds to induction BCG, a clinician should consider maintenance BCG for one year, as tolerated. (*Moderate Recommendation; Evidence Strength: Grade C*)
21. In a high-risk patient who completely responds to induction BCG, a clinician should continue maintenance BCG, based on availability, for three years, as tolerated. (*Moderate Recommendation; Evidence Strength: Grade B*)

BCG Relapse and Salvage Regimens

22. In an intermediate- or high-risk patient with persistent or recurrent disease or positive cytology following intravesical therapy, a clinician should consider performing prostatic urethral biopsy and an upper tract evaluation prior to administration of additional intravesical therapy. (*Conditional Recommendation; Evidence Strength: Grade C*)

23. In an intermediate- or high-risk patient with persistent or recurrent Ta or CIS disease after a single course of induction intravesical BCG, a clinician should offer a second course of BCG. (*Moderate Recommendation; Evidence Strength: Grade C*)
24. In a patient fit for surgery with high-grade T1 disease after a single course of induction intravesical BCG, a clinician should offer radical cystectomy. (*Moderate Recommendation; Evidence Strength: Grade C*)
25. A clinician should not prescribe additional BCG to a patient who is intolerant of BCG or has documented recurrence on TURBT of high-grade, non-muscle-invasive disease and/or CIS within six months of two induction courses of BCG or induction BCG plus maintenance. (*Moderate Recommendation; Evidence Strength: Grade C*)
26. In a patient with persistent or recurrent high-grade NMIBC within 12 months of completion of adequate BCG therapy (two induction courses or one induction course plus one maintenance cycle) who is unwilling or unfit for cystectomy, a clinician may recommend clinical trial enrollment, an alternative intravesical therapy (i.e., nadofaragene [firadenovec-vncg]) or alternative intravesical chemotherapies (gemcitabine/docetaxel). A clinician may also offer systemic immunotherapy with pembrolizumab to a patient with CIS within 12 months of completion of adequate BCG therapy. (*Conditional Recommendation; Evidence Strength: Grade C*)

Role of Cystectomy in NMIBC

27. In a patient with Ta low- or intermediate-risk disease, a clinician should not perform radical cystectomy until bladder-sparing modalities (staged TURBT, intravesical therapies) have failed. (*Clinical Principle*)
28. In a high-risk patient who is fit for surgery with persistent high-grade T1 disease on repeat resection, or T1 tumors with associated CIS, LVI, or variant histologies, a clinician should consider offering initial radical cystectomy. (*Moderate Recommendation; Evidence Strength: Grade C*)
29. In a high-risk patient with persistent or recurrent disease within one year following treatment with two induction cycles of BCG or BCG maintenance, a clinician should offer radical cystectomy. (*Moderate Recommendation; Evidence Strength: Grade C*)

Enhanced Cystoscopy

30. In a patient with NMIBC, a clinician should offer BLC at the time of TURBT, if available, to increase detection and decrease recurrence. (*Moderate Recommendation; Evidence Strength: Grade B*)
31. In a patient with NMIBC, a clinician may consider use of narrow-band imaging (NBI) to increase detection and decrease recurrence. (*Conditional Recommendation; Evidence Strength: Grade C*)

Risk Adjusted Surveillance and Follow-Up Strategies

32. After completion of the initial evaluation and treatment of a patient with NMIBC, a clinician should perform the first surveillance cystoscopy within three to four months. (*Expert Opinion*)
33. For a low-risk patient whose first surveillance cystoscopy is negative for tumor, a clinician should perform subsequent surveillance cystoscopy six to nine months later, and then annually thereafter; surveillance after five years in the absence of recurrence should be based on shared-decision making between the patient and clinician. (*Moderate Recommendation; Evidence Strength: Grade C*)
34. In an asymptomatic patient with a history of low-risk NMIBC, a clinician should not perform routine surveillance upper tract imaging. (*Expert Opinion*)
35. In a patient with a history of low-grade Ta disease and a noted sub-centimeter papillary tumor(s), a clinician may consider in-office fulguration as an alternative to resection under anesthesia. (*Expert Opinion*)
36. For an intermediate-risk patient whose first surveillance cystoscopy is negative for tumor, a clinician should perform subsequent cystoscopy with cytology every 3-6 months for 2 years, then 6-12 months for years 3 and 4, and then annually thereafter. (*Expert Opinion*)
37. For a high-risk patient whose first surveillance cystoscopy is negative for tumor, a clinician should perform subsequent cystoscopy with cytology every three to four months for two years, then six months for years three and four, and then annually thereafter. (*Expert Opinion*)
38. For an intermediate- or high-risk patient, a clinician should consider performing surveillance upper tract imaging at one- to two-year intervals. (*Expert Opinion*)

INTRODUCTION

Purpose

The survival rate for the majority of patients with NMIBC is favorable; however, the rates of recurrence and progression to muscle-invasive bladder cancer (MIBC) are important surrogate endpoints for overall prognosis, as these are major determinants of long-term outcome. The recurrence and progression probability rates depend on several clinical and pathologic factors. Therefore, the ability to predict risk of recurrence and progression and treat the disease appropriately is important. This guideline provides a risk-stratified clinical framework for the management of NMIBC.

Methodology

SYSTEMATIC REVIEW

The systematic review utilized in the creation of this guideline was completed in part through AHRQ and through additional supplementation that further addressed additional key questions and more recently published literature. A research librarian experienced in conducting literature searches for comparative effectiveness reviews searched in Ovid MEDLINE (January 1990 – October 2014), Cochrane Central Register of Controlled Trials (through September 2014), Cochrane Database of Systematic Reviews (through September 2014), Health Technology Assessment (through 3rd Quarter, 2014), National Health Sciences Economic Evaluation Database (through 3rd Quarter, 2014), and Database of Abstracts of Reviews of Effects (through 3rd Quarter, 2014) to capture both published and grey literature. Database searches resulted in 3,740 potentially relevant articles. After dual review of abstracts and titles, 643 articles were selected for full-text dual review, and 149 studies (in 192 publications) were determined to meet inclusion criteria and were included in this review. The AHRQ review was then updated by a consultant methodologist through September 2, 2015. Reference lists and previous systematic reviews were also reviewed for additional studies. This supplementation added 29 studies to the completed systematic review used in the creation of guideline statements.

In 2020, the NMIBC guideline was updated through the AUA amendment process in which newly published literature is reviewed and integrated into previously published guidelines in an effort to maintain currency. The amendment allowed for the incorporation of additional literature released since the initial publication of this guideline in 2016. For this updated literature review the methodology team searched Ovid MEDLINE[®] ALL from June 1, 2015 to November 22, 2019, and eliminated duplicate abstracts reviewed for earlier reports. Following initial report review, the Panel suggested additional abstracts that were assessed for inclusion as well. In total, the updated literature search identified 1,626 abstracts, of which 76 met inclusion criteria.

An additional update was performed in 2023. The updated search gathered literature from July 2019 to May 2023. This review identified 1,918 abstracts, of which 75 met inclusion criteria.

DATA EXTRACTION AND DATA MANAGEMENT

For treatment studies, the following information was extracted into evidence tables: study design, setting, inclusion and exclusion criteria, dose and duration of treatment for experimental and control groups, duration of follow-up, number of subjects screened, eligible and enrolled population characteristics (including age, race, sex, stage of disease, and functional status), results, adverse events, withdrawals due to adverse events, and sources of funding. Relative risks and associated 95 percent confidence intervals (CI) were calculated based on the information provided (sample sizes and incidence of outcomes in each intervention group). Discrepancies between calculated and reported results were noted when present.

For diagnostic accuracy studies, the following information was abstracted: setting, screening test or tests, method of data collection, reference standard, inclusion criteria, population characteristics (including age, sex, race, smoking status, signs or symptoms, and prior bladder cancer stage or grade), proportion of individuals with bladder cancer, bladder cancer stage and grade, definition of a positive screening exam, proportion of individuals unexaminable by the screening test, proportion who did not undergo reference standard, results, and sources of funding. When possible, two-by-two tables were created from information provided (sample size, prevalence, sensitivity, and specificity) and compared to calculated measures of diagnostic accuracy based on the two-by-two tables with reported results. Discrepancies between calculated and reported results were noted when present. Data extraction for each study was completed by one investigator and independently reviewed for accuracy and completeness by a second investigator.

ASSESSMENT OF THE RISK OF BIAS OF INDIVIDUAL STUDIES

Risk of bias was assessed for randomized trials and observational studies using criteria adapted from those developed by the U.S. Preventive Services Task Force.¹ Studies of diagnostic accuracy were rated using criteria adapted from QUADAS-2.² These criteria were applied in conjunction with the approaches recommended in the AHRQ Methods Guide³ for medical interventions and the AHRQ Methods Guide for Medical Test Reviews.⁴ Two investigators independently assessed the risk of bias of each study. Discrepancies were resolved through discussion and consensus. Each study was rated as “low,” “medium,” or “high” risk of bias.³

DETERMINATION OF EVIDENCE STRENGTH

The categorization of evidence strength is conceptually distinct from the quality of individual studies. Evidence strength refers to the body of evidence available for a particular question and includes not only individual study quality but consideration of study design, consistency of findings across studies, adequacy of sample sizes, and generalizability of samples, settings, and treatments for the purposes of the guideline. The AUA categorizes body of evidence strength as Grade A (well-conducted and highly-generalizable RCTs or exceptionally strong observational studies with consistent findings), Grade B (RCTs with some weaknesses of procedure or generalizability or moderately strong observational studies with consistent findings), or Grade C (RCTs with serious deficiencies of procedure or generalizability or extremely small sample sizes or observational studies that are inconsistent, have small sample sizes, or have other problems that potentially confound interpretation of data). By definition, Grade A evidence is evidence about which the Panel has a high level of certainty, Grade B evidence is evidence about which the Panel has a moderate level of certainty, and Grade C evidence is evidence about which the Panel has a low level of certainty (Table 1).⁵ The 38 statements created vary in level of evidence, but none include Level A evidence, and a majority are Level C evidence.

TABLE 1: Strength of Evidence Definitions		
AUA Strength of Evidence Category	GRADE Certainty Rating	Definition
A	High	• Very confident that the true effect lies close to that of the estimate of the effect
B	Moderate	• Moderately confident in the effect estimate • The true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different
C	Low	• Confidence in the effect estimate is limited • The true effect may be substantially different from the estimate of the effect
	Very Low	• Very little confidence in the effect estimate • The true effect is likely to be substantially different from the estimate of effect

AUA NOMENCLATURE: LINKING STATEMENT TYPE TO EVIDENCE STRENGTH

The AUA nomenclature system explicitly links statement type to body of evidence strength, level of certainty, magnitude of benefit or risk/burdens, and the Panel’s judgment regarding the balance between benefits and risks/burdens (Table 2). **Strong Recommendations** are directive statements that an action should (benefits outweigh risks/burdens) or should not (risks/burdens outweigh benefits) be undertaken because net benefit or net harm is substantial. **Moderate Recommendations** are directive statements that an action should (benefits outweigh risks/burdens) or should not (risks/burdens outweigh benefits) be undertaken because net benefit or net harm is moderate. **Conditional Recommendations** are non-directive statements used when the evidence indicates that there is no apparent net benefit or harm or when the balance between benefits and risks/burden is unclear. All three statement types may be supported by any body of evidence strength grade. Body of evidence strength Grade A in support of a Strong or Moderate Recommendation indicates that the statement can be applied to most patients in most circumstances and that future research is *unlikely to change confidence*. Body of evidence strength Grade B in support of a Strong or Moderate Recommendation indicates that the statement can be applied to most patients in most circumstances but that better evidence *could change confidence*. Body of evidence strength Grade C in support of a Strong or Moderate Recommendation indicates that the statement can be applied to most patients in most circumstances but that better evidence *is likely to change confidence*. Body of evidence strength Grade C is only rarely used in support of a Strong Recommendation. Conditional Recommendations also can be supported by any evidence strength. When body of evidence strength is Grade A, the statement indicates that benefits and risks/burdens appear balanced, the best action depends on patient circumstances, and future research is *unlikely to change confidence*. When body of evidence strength Grade B is used, benefits and risks/burdens appear balanced, the best action also depends on individual patient circumstances and better evidence *could change confidence*. When body of evidence strength Grade C is used, there is uncertainty regarding the balance between benefits and risks/burdens, alternative strategies may be equally reasonable, and better evidence is *likely to change confidence*.

Where gaps in the evidence existed, the Panel provides guidance in the form of *Clinical Principles* or *Expert Opinion* with consensus achieved using a modified Delphi technique if differences of opinion emerged.⁶ A *Clinical Principle* is a statement about a component of clinical care that is widely agreed upon by urologists or other clinicians for which there may or may not be evidence in the medical literature. *Expert Opinion* refers to a statement, achieved by consensus of the Panel, which is based on members’ clinical training, experience, knowledge, and judgment for which there is no evidence.

TABLE 2: AUA Nomenclature Linking Statement Type to Level of Certainty, Magnitude of Benefit or Risk/Burden, and Body of Evidence Strength			
Evidence Grade	Evidence Strength A (High Certainty)	Evidence Strength B (Moderate Certainty)	Evidence Strength C (Low Certainty)
Strong Recommendation (Net benefit or harm substantial)	-Benefits > Risks/Burdens (or vice versa) -Net benefit (or net harm) is substantial -Applies to most patients in most circumstances and future research is unlikely to change confidence	-Benefits > Risks/Burdens (or vice versa) -Net benefit (or net harm) is substantial -Applies to most patients in most circumstances but better evidence could change confidence	-Benefits > Risks/Burdens (or vice versa) -Net benefit (or net harm) appears substantial -Applies to most patients in most circumstances but better evidence is likely to change confidence (rarely used to support a Strong Recommendation)
Moderate Recommendation (Net benefit or harm moderate)	-Benefits > Risks/Burdens (or vice versa) -Net benefit (or net harm) is moderate -Applies to most patients in most circumstances and future research is unlikely to change confidence	-Benefits > Risks/Burdens (or vice versa) -Net benefit (or net harm) is moderate -Applies to most patients in most circumstances but better evidence could change confidence	-Benefits > Risks/Burdens (or vice versa) -Net benefit (or net harm) appears moderate -Applies to most patients in most circumstances but better evidence is likely to change confidence
Conditional Recommendation (Net benefit or harm comparable to other options)	-Benefits = Risks/Burdens -Best action depends on individual patient circumstances -Future Research is unlikely to change confidence	-Benefits = Risks/Burdens -Best action appears to depend on individual patient circumstances -Better evidence could change confidence	-Balance between Benefits & Risks/Burdens unclear -Net benefit (or net harm) comparable to other options -Alternative strategies may be equally reasonable -Better evidence likely to change confidence
Clinical Principle	a statement about a component of clinical care that is widely agreed upon by urologists or other clinicians for which there may or may not be evidence in the medical literature		
Expert Opinion	a statement, achieved by consensus of the Panel, that is based on members' clinical training, experience, knowledge, and judgment for which there may or may not be evidence in the medical literature		

PROCESS

The NMIBC Panel was created in 2014 by the American Urological Association Education and Research, Inc. (AUA). The Practice Guidelines Committee (PGC) of the AUA selected the Panel Chair who in turn appointed the Vice Chair. In a collaborative process, additional Panel members, including additional members of the Society of Urologic Oncology (SUO) with specific expertise in this area, were then nominated and approved by the PGC. The AUA conducted a thorough peer review process. The draft guidelines document was distributed to 128 peer reviewers, 66 of which submitted comments. The panel reviewed and discussed all submitted comments and revised the draft as needed. Once finalized, the guideline was submitted for approval to the PGC and Science and Quality Council (S&Q). Then it was submitted to the AUA Board of Directors for final approval.

The 2020 amendment also underwent peer review. The draft amendment was distributed to 77 peer reviewers, 21 of whom submitted 57 comments. The Panel reviewed and discussed all submitted comments and revised the draft as needed. Once finalized, the amendment was submitted for approval in the same manner as with the full guideline.

Additionally, the 2024 amendment underwent peer review. The draft amendment was distributed to 83 peer reviewers, 18 of whom submitted 80 comments. The Panel reviewed and discussed all submitted comments and revised the draft as needed. Once finalized, the amendment was submitted for approval in the same manner as with the full guideline.

Funding of the Panel was provided by the AUA; Panel members received no remuneration for their work.

BACKGROUND

Epidemiology

NMIBC represents approximately 75% of the 74,000 estimated new bladder cancer cases diagnosed in the United States in 2015.^{7, 8} Bladder cancer is more common in males than females with a ratio of approximately 3:1, and it is the fourth most common solid malignancy in men. There are 16,000 estimated deaths for 2015, predominantly affecting males.^{7, 9} Bladder cancer primarily affects Caucasian Americans and those older than 65 years with relatively stable mortality rates since 1975.⁹

National registry data from the U.S. Surveillance Epidemiology and End Results program demonstrates that the incidence of all stages of NMIBC has been relatively stable from 1988-2006; however, the adjusted incidence of stage Ta has significantly increased, while stages Tis and T1 have slightly decreased.¹⁰

Etiology

RISK FACTORS

Multiple factors are associated with bladder carcinogenesis; however, tobacco smoking is the most significant and most common risk factor.¹¹ Although smoking cessation may somewhat decrease carcinogenesis risk, former smokers still have a higher risk of bladder cancer than those who never smoked.¹¹ With respect to NMIBC, current tobacco use and cumulative lifetime exposure may be associated with recurrence and progression.^{12, 13} Although an incomplete list, the Panel has identified other more common risk factors. Occupational exposure to chemical carcinogens, such as aromatic amines, polycyclic aromatic hydrocarbons, and arsenic, is another reported risk factor.^{14, 15} Patients with other malignancies, such as lymphomas and leukemias, who receive treatment with cyclophosphamide may be at increased risk for bladder cancer.^{16, 17} Patients with Lynch Syndrome may also be at increased risk of urothelial carcinoma of the bladder, as well as, the upper urinary tract.^{18, 19} Infection also increases the risk of bladder cancer; in particular, *Schistosoma hematobium*, the pathogen responsible for schistosomiasis, is a risk factor for squamous cell carcinoma of the bladder in certain regions of the world.²⁰ In looking at squamous cell carcinoma of the bladder, chronic catheter use also serves as a risk factor. Additionally, aristolochic acid, a natural compound found in a number of plants of the *Aristolochia* genus, has been linked to upper-tract urothelial carcinoma.²¹ Another known risk factor includes external beam radiation to the pelvis.²²

MOLECULAR MECHANISM AND GENETICS

There is no currently accepted genetic or inheritable cause of bladder cancer; however, studies suggest that genomic instability and genetic pathway mutations/alterations may play a role in bladder carcinogenesis. Studies suggest that polymorphisms in two carcinogen-detoxifying genes GSTM-1 and NAT-2 may be responsible for increased susceptibility to developing bladder cancer in certain patients.²³ Chromosome 9 deletion is a common genetic alteration found in NMIBC, with loss of heterozygosity (LOH) of 9p, homozygous deletion of CDKN2A, and loss of expression of p16 in NMIBC predicting recurrence free survival.²⁴⁻²⁶ Mutations in tumor suppressor genes can lead to disruption of cell cycle regulation and predispose to carcinogenesis. CIS frequently demonstrates mutations in the tumor suppressor genes TP53, RB1 (retinoblastoma), and PTEN.²⁷ Oncogenes that promote tumor cell development and alterations in FGFR3, PIK3CA, and RAS are common in NMIBC.^{27, 28}

Presentation and Diagnosis

The most common presenting symptom is painless hematuria (gross or microscopic). According to the AUA Guideline on the diagnosis, evaluation, and follow-up of patients with asymptomatic microhematuria (AMH), the rate of urinary tract malignancy in AMH is approximately 2.6%.²⁹ Irritative voiding symptoms (e.g., frequency, urgency, dysuria) may also be associated with CIS in patients with no sign of urinary tract infection (UTI). Physical exam rarely reveals significant findings in patients with NMIBC. However, a bimanual exam may be performed under anesthesia at the time of TURBT and should be performed at that time if the tumor appears invasive. Although not indicated for routine screening and evaluation of AMH, urinary cytology (voided or barbotage) may be used in the surveillance of bladder cancer for certain patients as it possesses a high sensitivity and positive predictive value for high-grade tumors and CIS.^{30, 31} Contrast-based axial imaging, such as computed tomography (CT) or magnetic resonance imaging (MRI) is the recommended imaging modality during the work-up for bladder cancer. Retrograde pyelogram and intravenous urography may also be used when CT or MRI are unavailable. Abdomino-pelvic sonography alone may not provide sufficient anatomic detail for upper urinary tract imaging during the work-up of bladder cancer.³²

The diagnosis of bladder cancer is confirmed by direct visualization of the tumor and other mucosal abnormalities with endoscopic excision using cystoscopy and TURBT. An adequate TURBT requires complete resection of all visible tumor with adequate sampling of the bladder to assess the depth of invasion.

Staging and Grading

Staging for bladder cancer is separated into clinical and pathologic stage, as outlined by the American Joint Committee on Cancer (AJCC), also known as the Tumor-Node-Metastases (TNM) classification.³³ Clinical stage reflects the histologic findings at TURBT; the clinician’s physical exam, including bimanual exam under anesthesia; and findings on radiologic imaging. The pathologic report of the TURBT should indicate whether lamina propria and muscularis propria are present as well as the degree of involvement, if present. In addition, effort should be made by the pathologist to examine the specimen for lymphovascular invasion (LVI), when applicable, as this is associated with worse prognosis.³⁴⁻³⁷ Pathological staging, also known as surgical staging, is based on the extent of disease following surgical resection of the bladder (partial versus radical cystectomy) and of the adjacent pelvic lymph nodes. Under the AJCC staging system, NMIBC includes the following: (1) papillary tumors confined to the epithelial mucosa (stage Ta), (2) tumors invading the subepithelial tissue (i.e., lamina propria; T1), and (3) Tis. (Table 3)

TABLE 3: Staging of primary tumors (T) in bladder cancer ³³	
TX	Primary tumor cannot be assessed
Ta	Noninvasive papillary carcinoma
Tis	Carcinoma in situ (CIS)
T1	Tumor invades lamina propria
T2	Tumor invades muscularis propria
T2a	Tumor invades superficial muscularis propria (inner half)
T2b	Tumor invades deep muscularis propria (outer half)
T3	Tumor invades perivesical tissue/fat
T3a	Tumor invades perivesical tissue/fat microscopically
T3b	Tumor invades perivesical tissue fat macroscopically (extravesical mass)
T4	Tumor invades prostate, uterus, vagina, pelvic wall, or abdominal wall
T4a	Tumor invades adjacent organs (uterus, ovaries, prostate stoma)
T4b	Tumor invades pelvic wall and/or abdominal wall

Tumor grade is an important prognostic factor for determining risk of recurrence and progression in bladder cancer. Prior to the 2004 revised classification, the 1973 World Health Organization (WHO) classification was the widely accepted format for grading bladder neoplasia.^{38, 39} The 1973 version designated tumors as either papilloma, grade 1, 2, or 3, whereas the 2004 revision designated tumors as ‘low’ or ‘high’ grade. The 1973 grade 2 or ‘intermediate’ grade tumors are now re-classified as either ‘low’ or ‘high’ grade depending on cellular morphology.^{38, 39} In addition, the 2004 classification introduced the new category of papillary urothelial neoplasm of low malignant potential to describe lesions with an increased number of urothelial layers when compared with papilloma but without cytologic features of malignancy. The WHO/International Society of Urological Pathology 2004 grading system is now the most widely accepted and utilized system in the United States. (Table 4)

TABLE 4: 2004 WHO/ International Society of Urologic Pathologists: Classification of Non-muscle Invasive Urothelial Neoplasia ³⁸
Hyperplasia (flat and papillary)
Reactive atypia
Atypia of unknown significance
Urothelial dysplasia
Urothelial CIS
Urothelial papilloma
Papillary urothelial neoplasm of low malignant potential
Non-muscle invasive low-grade papillary urothelial carcinoma
Non-muscle invasive high-grade papillary urothelial carcinoma

Prognosis

The survival prognosis for patients with NMIBC is relatively favorable, with the cancer-specific survival (CSS) in high-grade disease ranging from approximately 70-85% at 10 years and a much higher rate for low-grade disease.^{40, 41} The rates of recurrence and progression to MIBC are important surrogate endpoints for prognosis in NMIBC, as these are major determinants of long-term outcome. However, NMIBC is a clinically heterogeneous group of cancers with a wide range of recurrence and progression probabilities that depend on several clinical and pathologic factors. For example, long-term follow up of low-grade Ta lesions demonstrates a recurrence rate of approximately 55%, but with a much lower percentage (6%) experiencing stage progression.⁴² In contrast, high-grade T1 lesions have both a significant risk of recurrence (45%) and increased chance of progression (17%) in single institution series.⁴⁰ Therefore, the ability to predict recurrence and progression risk in NMIBC, based on patient-specific disease characteristics, holds prognostic significance. Risk stratification in NMIBC aids personalized treatment decisions and surveillance strategies as opposed to a generalized 'one-size fits all' approach.

Risk Stratification

Significant effort has been put forth to develop tools for risk stratification and prognostication. A widely published system is the European Organization for Research and Treatment of Cancer (EORTC) risk calculator, based on the combined data from seven trials involving patients with NMIBC.⁴³ Using clinical and pathologic variables in a scoring system, the EORTC calculator provides a probability of recurrence and progression at one and five years. Important factors for recurrence identified by the EORTC study include prior recurrence rate, number of tumors, and tumor size.⁴³ With respect to progression, important factors include T-stage, presence of CIS, and grade. A second risk stratification tool is that developed by the Spanish Urological Club for Oncological Treatment/Club Urologico Espanol de Tratamiento Oncologico (CUETO).⁴⁴ These models are examples of carefully constructed risk stratification systems; however, they have limitations. Both tools are limited by lack of applicability to current patient populations because few patients from the development cohort received BCG maintenance, underwent re-staging transurethral-resection, or received single-dose post-operative mitomycin C. A recent update of the EORTC nomogram for risk stratification attempted to address the lack of BCG maintenance in prior studies, by analyzing a cohort of patients treated with one to three years of BCG. This updated study cohort lacked patients with CIS and again was limited by absence of routine re-resection.⁴⁵ Additionally, the EORTC risk calculator utilizes the 1973 WHO grading system to generate risk probabilities as opposed to the 2004 version. As previously mentioned, the 2004 revision is the currently accepted classification for tumor grade; therefore, the EORTC risk tables are commonly not considered in the U.S.

Risk groupings are evaluated by their ability to predict the outcome of patients who are felt to be similar to one another. The most commonly used tool to assess the accuracy of risk groupings is the concordance index (C-index). The C-index is a measure of the ability of a risk assessment tool to separate those patients with the outcome of interest from those without the outcome of interest (e.g., recurrence or progression).⁴⁶ A C-index of 0.5 implies that the ability to predict outcome is no better than random chance. For the original EORTC study, the C-indices for recurrence and progression were 0.66 and 0.75, respectively and 0.64 and 0.7, respectively, for the CUETO study.^{43, 44} A further important limitation of the existing risk stratification models is that neither reported formal measures of calibration (the degree to which predicted and observed risk estimates agree). Several studies have retrospectively evaluated the ability of the EORTC and CUETO models to predict the risk of recurrence and progression in other patient populations.⁴⁷⁻⁵² These attempts at external validation using other patient populations have yielded variable C-index results and underscore the fact that both instruments are limited by suboptimal calibration and inherent biases based on their designs. Evaluation of these studies investigating the utility of risk stratification in multiple populations demonstrate that they have a poor to fair ability to discriminate risk of recurrence (C-index 0.52 to 0.66) and good to fair (C-index 0.62 to 0.81) ability to discriminate risk of progression.

The Panel acknowledges that Level A evidence does not support stratification as affecting disease recurrence, progression, or survival. However, despite the lack of evidence confirming a positive influence on clinical outcome, the Panel agrees that there is value to creating fundamental categories that broadly estimate the likelihood of recurrence and progression. The Panel set out to create such a system, with categories summarized as 'low,' 'intermediate,' and 'high' risk for recurrence and/or progression. (**Table 5**) This risk grouping system is a simple tool, intended for use in clinical practice as a general framework for guiding patient counseling and aiding in treatment and surveillance decisions based on prognosis. While there are similarities between the current risk categories outlined in the Guideline and the EORTC stratification, it should be noted that they are not based on a meta-analysis or original studies and represent the Panel's consensus regarding the likelihood of recurrence and progression. To develop the current risk groupings, the Panel set forth defining first those at lowest and highest risk for recurrence and/or progression. Numerous clinical scenarios based on disease characteristics were then incorporated into the grouping system, and each one was placed into a category based on unanimous expert consensus and available published data. The Panel also recognizes that the intermediate group is somewhat heterogeneous, and the outcome of patients within this group may still exhibit some variation along the spectrum of risk of recurrence and progression.

Unique to the AUA/SUO Guideline Risk Stratification System is the incorporation of prior BCG intravesical therapy on prognosis. There are limited data that demonstrate that patients who have persistent or recurrent disease at six months following BCG therapy are at increased risk of disease progression.^{53, 54} As such, the Panel reasons that patients who are intermediate risk and demonstrate BCG failure should be re-stratified to the high-risk group. The rationale for this approach is that those patients who do not respond to standard intravesical therapy likely harbor more aggressive disease than implied by clinical or pathologic features; therefore, a lack of response serves as a surrogate marker for increased risk of recurrence and/or progression. The Panel also understands and appreciates that within each of these risk strata that an individual patient may have more or less concerning features that can influence care.

The Panel acknowledges the need for validation of these risk groups in large, contemporary patient cohorts in order to assess the model’s performance for predicting disease recurrence and progression.

Relevance of the International BCG Shortage to the AUA Guidelines

The global shortages in TICE BCG that occurred in 2014 and 2019 led the AUA to recommend several management strategies to maintain high quality care for patients with NMIBC. These recommendations may supersede the guideline statements below. In particular, the BCG shortage impacts guideline statements 17, 20, and 21. The AUA Statement on the BCG Shortage is available at <https://www.auanet.org/about-us/bcg-shortage-info>.

TABLE 5: AUA Risk Stratification for NMIBC		
Low Risk	Intermediate Risk	High Risk
LG ^a solitary Ta ≤ 3cm	Recurrence within 1 year, LG Ta	HG T1
PUNLMP ^b	Solitary LG Ta > 3cm	Any recurrent, HG Ta
	LG Ta, multifocal	HG Ta, >3cm (or multifocal)
	HG ^c Ta, ≤ 3cm	Any CIS ^d
	LG T1	Any BCG failure in HG patient
		Any variant histology
		Any LVI ^e
		Any HG prostatic urethral involvement
^a LG = low grade; ^b PUNLMP = papillary urothelial neoplasm of low malignant potential; ^c HG = high grade; ^d CIS=carcinoma <i>in situ</i> ; ^e LVI = lymphovascular invasion		

GUIDELINE STATEMENTS

Diagnosis

GUIDELINE STATEMENT 1

At the time of resection of suspected bladder cancer, a clinician should perform a thorough cystoscopic examination of a patient’s entire urethra and bladder that evaluates and documents tumor size, location, configuration, number, and mucosal abnormalities. **(Clinical Principle)**

Discussion

GUIDELINE STATEMENT 2

At initial diagnosis of a patient with bladder cancer, a clinician should perform complete visual resection of the bladder tumor(s), when technically feasible. **(Clinical Principle)**

Discussion

GUIDELINE STATEMENT 3

A clinician should perform upper urinary tract imaging as a component of the initial evaluation of a patient with bladder cancer. *(Clinical Principle)*

Discussion

GUIDELINE STATEMENT 4

In a patient with a history of NMIBC with normal cystoscopy and positive cytology, a clinician should consider prostatic urethral biopsies and upper tract imaging, as well as enhanced cystoscopic techniques (blue light cystoscopy [BLC], when available), ureteroscopy, or random bladder biopsies. *(Expert Opinion)*

Discussion

Risk Stratification

GUIDELINE STATEMENT 5

At the time of each occurrence/recurrence, a clinician should assign a clinical stage and classify a patient accordingly as “low-,” “intermediate-,” or “high-risk.” *(Moderate Recommendation; Evidence Strength: Grade C)*

Discussion

Variant Histologies

GUIDELINE STATEMENT 6

An experienced genitourinary pathologist should review the pathology of a patient with any doubt in regard to variant or suspected variant histology (e.g., micropapillary, nested, plasmacytoid, neuroendocrine, sarcomatoid), extensive squamous or glandular differentiation, or the presence/absence of lymphovascular invasion (LVI). *(Moderate Recommendation; Evidence Strength: Grade C)*

Discussion

GUIDELINE STATEMENT 7

If a bladder sparing approach is being considered in a patient with variant histology, then a clinician should perform a restaging transurethral resection of bladder tumor (TURBT) within four to six weeks of the initial TURBT. *(Expert Opinion)*

Discussion

GUIDELINE STATEMENT 8

Due to the high rate of upstaging associated with variant histology, a clinician should consider offering initial radical cystectomy. *(Expert Opinion)*

Discussion

Urine Markers after Diagnosis of Bladder Cancer

GUIDELINE STATEMENT 9

In surveillance of NMIBC, a clinician should not use urinary biomarkers in place of cystoscopic evaluation. *(Strong Recommendation; Evidence Strength: Grade B)*

Discussion

GUIDELINE STATEMENT 10

In a patient with a history of low-risk cancer and a normal cystoscopy, a clinician should not routinely use a urinary biomarker or cytology during surveillance. *(Expert Opinion)*

Discussion

GUIDELINE STATEMENT 11

In a patient with NMIBC, a clinician may use biomarkers to assess response to intravesical BCG (UroVysion® FISH) and adjudicate equivocal cytology (UroVysion® FISH and ImmunoCyt™). *(Expert Opinion)*

Discussion

TURBT/ Repeat Resection: Timing, Technique, Goal, Indication

GUIDELINE STATEMENT 12

In a patient with non-muscle invasive disease who underwent an incomplete initial resection (not all visible tumor treated), a clinician should perform repeat transurethral resection or endoscopic treatment of all remaining tumor if technically feasible. *(Strong Recommendation; Evidence Strength: Grade B)*

Discussion

GUIDELINE STATEMENT 13

In a patient with high-risk, high-grade Ta tumors, a clinician should consider performing repeat transurethral resection of the primary tumor site within six weeks of the initial TURBT. *(Moderate Recommendation; Evidence Strength: Grade C)*

Discussion

GUIDELINE STATEMENT 14

In a patient with T1 disease, a clinician should perform repeat transurethral resection of the primary tumor site to include muscularis propria within six weeks of the initial TURBT. *(Strong Recommendation; Evidence Strength: Grade B)*

Discussion

Intravesical Therapy; BCG/Maintenance; Chemotherapy/BCG Combinations

GUIDELINE STATEMENT 15

In a patient with suspected or known low- or intermediate-risk bladder cancer, a clinician should consider administration of a single postoperative instillation of intravesical chemotherapy (e.g., gemcitabine, mitomycin C) within 24 hours of TURBT. In a patient with a suspected perforation or extensive resection, a clinician should not use postoperative intravesical chemotherapy. *(Moderate Recommendation; Evidence Strength: Grade B)*

Discussion

GUIDELINE STATEMENT 16

In a low-risk patient, a clinician should not administer induction intravesical therapy. *(Moderate Recommendation; Evidence Strength: Grade C)*

Discussion

GUIDELINE STATEMENT 17

In an intermediate-risk patient a clinician should consider administration of a six-week course of induction intravesical chemotherapy or immunotherapy. *(Moderate Recommendation; Evidence Strength: Grade B)*

Discussion

GUIDELINE STATEMENT 18

In a high-risk patient with newly diagnosed carcinoma *in situ* (CIS), high-grade T1, or high-risk Ta urothelial carcinoma, a clinician should administer a six-week induction course of BCG. *(Strong Recommendation; Evidence Strength: Grade B)*

Discussion

GUIDELINE STATEMENT 19

In an intermediate-risk patient who completely responds to an induction course of intravesical chemotherapy, a clinician may utilize maintenance therapy. *(Conditional Recommendation; Evidence Strength: Grade C)*

Discussion

GUIDELINE STATEMENT 20

In an intermediate-risk patient who completely responds to induction BCG, a clinician should consider maintenance BCG for one year, as tolerated. *(Moderate Recommendation; Evidence Strength: Grade C)*

Discussion

GUIDELINE STATEMENT 21

In a high-risk patient who completely responds to induction BCG, a clinician should continue maintenance BCG, based on availability, for three years, as tolerated. *(Moderate Recommendation; Evidence Strength: Grade B)*

Discussion

BCG Relapse and Salvage Regimens

GUIDELINE STATEMENT 22

In an intermediate- or high-risk patient with persistent or recurrent disease or positive cytology following intravesical therapy, a clinician should consider performing prostatic urethral biopsy and an upper tract evaluation prior to administration of additional intravesical therapy. *(Conditional Recommendation; Evidence Strength: Grade C)*

Discussion

GUIDELINE STATEMENT 23

In an intermediate- or high-risk patient with persistent or recurrent Ta or CIS disease after a single course of induction intravesical BCG, a clinician should offer a second course of BCG. *(Moderate Recommendation; Evidence Strength: Grade C)*

Discussion

GUIDELINE STATEMENT 24

In a patient fit for surgery with high-grade T1 disease after a single course of induction intravesical BCG, a clinician should offer radical cystectomy. *(Moderate Recommendation; Evidence Strength: Grade C)*

Discussion

GUIDELINE STATEMENT 25

A clinician should not prescribe additional BCG to a patient who is intolerant of BCG or has documented recurrence on TURBT of high-grade, non-muscle-invasive disease and/or CIS within six months of two induction courses of BCG or induction BCG plus maintenance. *(Moderate Recommendation; Evidence Strength: Grade C)*

Discussion

GUIDELINE STATEMENT 26

In a patient with persistent or recurrent high-grade NMIBC within 12 months of completion of adequate BCG therapy (two induction courses or one induction course plus one maintenance cycle) who is unwilling or unfit for cystectomy following two courses of BCG, a clinician may recommend clinical trial enrollment, an alternative intravesical therapy (i.e., nadofaragene [firadenovec-vncg]) or alternative intravesical chemotherapies (gemcitabine/docetaxel). A clinician may also offer systemic immunotherapy with pembrolizumab to a patient with CIS within 12 months of completion of adequate BCG therapy. *(Conditional Recommendation; Evidence Strength: Grade C)*

Discussion

Role of Cystectomy in NMIBC

GUIDELINE STATEMENT 27

In a patient with Ta low- or intermediate-risk disease, a clinician should not perform radical cystectomy until bladder-sparing modalities (staged TURBT, intravesical therapies) have failed. *(Clinical Principle)*

Discussion

GUIDELINE STATEMENT 28

In a high-risk patient who is fit for surgery with persistent high-grade T1 disease on repeat resection, or T1 tumors with associated CIS, LVI, or variant histologies, a clinician should consider offering initial radical cystectomy. *(Moderate Recommendation; Evidence Strength: Grade C)*

Discussion

GUIDELINE STATEMENT 29

In a high-risk patient with persistent or recurrent disease within one year following treatment with two induction cycles of BCG or BCG maintenance, a clinician should offer radical cystectomy. *(Moderate Recommendation; Evidence Strength: Grade C)*

Discussion

Enhanced Cystoscopy

GUIDELINE STATEMENT 30

In a patient with NMIBC, a clinician should offer BLC at the time of TURBT, if available, to increase detection and decrease recurrence. *(Moderate Recommendation; Evidence Strength: Grade B)*

Discussion

GUIDELINE STATEMENT 31

In a patient with NMIBC, a clinician may consider use of narrow-band imaging (NBI) to increase detection and decrease recurrence. *(Conditional Recommendation; Evidence Strength: Grade C)*

Discussion

Risk Adjusted Surveillance and Follow-up Strategies

GUIDELINE STATEMENT 32

After completion of the initial evaluation and treatment of a patient with NMIBC, a clinician should perform the first surveillance cystoscopy within three to four months. *(Expert Opinion)*

Discussion

GUIDELINE STATEMENT 33

For a low-risk patient whose first surveillance cystoscopy is negative for tumor, a clinician should perform subsequent surveillance cystoscopy six to nine months later, and then annually thereafter; surveillance after five years in the absence of recurrence should be based on shared-decision making between the patient and clinician. *(Moderate Recommendation; Evidence Strength: Grade C)*

Discussion

GUIDELINE STATEMENT 34

In an asymptomatic patient with a history of low-risk NMIBC, a clinician should not perform routine surveillance upper tract imaging. *(Expert Opinion)*

Discussion

GUIDELINE STATEMENT 35

In a patient with a history of low-grade Ta disease and a noted sub-centimeter papillary tumor(s), a clinician may consider in-office fulguration as an alternative to resection under anesthesia. *(Expert Opinion)*

Discussion

GUIDELINE STATEMENT 36

For an intermediate-risk patient whose first surveillance cystoscopy is negative for tumor, a clinician should perform subsequent cystoscopy with cytology every 3-6 months for 2 years, then 6-12 months for years 3 and 4, and then annually thereafter. *(Expert Opinion)*

Discussion

GUIDELINE STATEMENT 37

For a high-risk patient whose first surveillance cystoscopy is negative for tumor, a clinician should perform subsequent cystoscopy with cytology every three to four months for two years, then six months for years three and four, and then annually thereafter. *(Expert Opinion)*

Discussion

GUIDELINE STATEMENT 38

For an intermediate- or high-risk patient, a clinician should consider performing surveillance upper tract imaging at one- to two- year intervals. *(Expert Opinion)*

Discussion

FUTURE DIRECTIONS

The future of NMIBC will likely be driven forward by basic science, novel technologies, new therapeutics and clinical trials. The bladder cancer genome atlas project provided analysis of 131 muscle-invasive urothelial carcinomas in an effort to describe molecular alterations and, ideally, provide insight into use of molecularly targeted agents for both muscle-invasive and NMIBC. The NMIBC community is fortunate to have a multitude of clinical trials currently in this disease space, the vast majority of which are studying novel agents to improve outcomes of BCG or treat BCG failures with both intravesical and systemic agents. There are also several trials investigating new technology, surgical techniques, radiation therapy, and variable surveillance schedules.

As new treatment alternatives for NMIBC are being examined, research should continue in optimizing the dosing, scheduling, and administration of currently used medications that have already shown efficacy.

Novel urinary biomarkers. Although the current consensus of the guideline panel describes a limited role for urinary biomarkers to replace cystoscopic surveillance in NMIBC, the future directions in this field hold promise. Advances in sensitivity for detection of high-grade disease in a surveillance population of high-grade NMIBC patients using the CX Bladder platform have been significant. In addition the recent review article by Rose and colleagues has outlined the future applications of urinary cell free DNA in both detection and molecular risk stratification of patients with NMIBC and the Panel believes that this technology holds promise for future clinical application.²⁷⁴

Novel agents to improve BCG efficacy or manage BCG failures. Management of patients with intermediate- or high-risk bladder cancer recurrences after two induction courses of BCG unwilling or unfit for radical cystectomy remains uncertain. Although many different salvage intravesical therapies have been evaluated, these studies are generally limited by small patient numbers, modest improvements in recurrence-free survival with respective intravesical agent(s), and no significant effects on progression or survival. These limitations highlight the dire need for novel agents in this disease setting. For instance, immune checkpoint inhibitors have been a resounding success in metastatic bladder cancer with trials already underway moving these agents into earlier stages of bladder cancer, including adjuvant, neoadjuvant and NMIBC settings.²⁷⁵ There are multiple current open clinical trials evaluating novel agents for BCG failures, including an oncolytic virus regimen (BOND 2), recombinant fusion proteins (Vicinium), immune modulation (ALT-801, HS-410, ALT-803, PANVAC), cytotoxic therapies (cabazitaxel, gemcitabine, cisplatin,) and targeted small molecule kinase inhibitors (sunitinib, dovitinib, erlotinib). In an open-label, multicenter, parallel-arm, phase II study, 43 patients with HG BCG-unresponsive or relapsed NMIBC received intravesical nadofaragene (firadenovec-vncg).²⁷⁶ Fourteen patients remained free of HG recurrence 12 months after initial treatment. These results lead to the FDA approval of this agent in 2022. More recently in the Quilt -3.0-32 trial, Chamie and colleagues reported their results using a combination of BCG and nogapendekin alfa inbakicept, an IL-15 superagonist.²⁷⁷ This combination therapy achieved a one-year disease free survival of 45% in BCG-unresponsive CIS and papillary bladder cancer with limited toxicity.

As research continues in this space, we are likely to see an increase in the number of available treatment options for such patients.

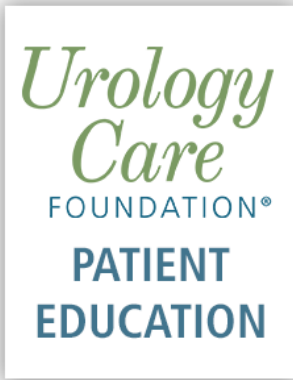
New technologies. Enhanced cystoscopy, including BLC at time of TURBT, has been demonstrated in multiple studies to decrease bladder tumor recurrence and seems particularly valuable in evaluation of positive urinary cytology in the setting of negative WLC.⁷³ Further studies of new technologies in management of patients with NMIBC include a current phase IV trial (NCT01567462) underway to evaluate TURBT using a PK button vaporization electrode compared to standard monopolar loop electrocautery. Investigators hypothesize TURBT using PK button vaporization may be less invasive with fewer side effects and improved patient recovery.

Therapeutic Trials in Surgery/Radiation. There are minimal data to support chemoradiation in the management of high-risk NMIBC,²⁵⁰ but current trials are underway evaluating the role of radiation therapy in patients with recurrent high-grade T1 disease after intravesical BCG. RTOG 0926 (NCT00981656) is a Phase II trial evaluating chemoradiation (cisplatin or mitomycin/fluorouracil with three-dimensional conformal radiation therapy) with a primary endpoint of three-year freedom from radical cystectomy and several secondary endpoints, including progression-free, disease-specific and overall survival.

Imaging. The advent of multiparametric MRI imaging has led to advances in the accuracy of staging both NMIBC and MIBC. In centers of expertise, the use of the vesical imaging reporting and data system (VI-RADS) coupled with state of the art 3 Tesla MR systems, has reported outstanding sensitivity and specificity for detection of MIBC in the setting of high-risk NMIBC.²⁷⁸ If reproducible, this form of imaging may lead to a decrease in the burden of re-TURBT and improved selection of patients with MIBC for more appropriate therapy.²⁷⁸

Surveillance. Finally, a randomized pilot clinical trial (NCT02298998) evaluating common surveillance schedules could significantly impact follow-up in patients with NMIBC. Patients will be randomized to either cystoscopy at 3 months, 12 months and then annually for 5 years versus cystoscopy every 3 months for 2 years, every 6 months for 2 years and annually thereafter. The primary objectives of this study include development of methodology to assess both patient satisfaction and costs associated with cystoscopy for bladder cancer surveillance with secondary objectives of cost, number of overall procedures and proportion of patients with disease recurrence and progression at two years. Monitoring patients with NMIBC less frequently may potentially decrease costs and improve patient satisfaction without increased risk of progression to muscle-invasive disease.

Tools and Resources



[Bladder Cancer Filtered Resources](#)

ABBREVIATIONS

AHRQ	Agency for Healthcare Research and Quality
AJCC	American Joint Committee on Cancer
AMH	Asymptomatic Microhematuria
BLC	Blue light cystoscopy
CSS	Cancer-specific survival
CIS	Carcinoma in situ
CT	Computed tomography

CUETO	Club Urologico Español de Tratamiento Oncologico
EORTC	European Organization for Research and Treatment of Cancer
LVI	Lymphovascular invasion
MRI	Magnetic resonance imaging
NBI	Narrow band imaging
NMIBC	Non-muscle invasive bladder cancer
US	Ultrasonography
TURBT	Transurethral resection of bladder tumor
UTI	Urinary tract infection
WHO	World Health Organization
WLC	White light cystoscopy

REFERENCES

1. U.S. Preventive services task force procedure manual: AHRQ Publication No. 08-05118- EF, 2008
2. Whiting PF, Rutjes AW, Westwood ME et al: Quadas-2: A revised tool for the quality assessment of diagnostic accuracy studies. Ann Intern Med 2011; **155**: 529.
3. Methods guide for effectiveness and comparative effectiveness reviews. Agency for healthcare research and quality. Rockville, MD, 2014
4. Ahrq methods for effective health care. In: Methods guide for medical test reviews. Edited by S. M. Chang, D. B. Matchar, G. W. Smetana, C. A. Umscheid. Rockville (MD): Agency for Healthcare Research and Quality (US), 2012
5. Faraday M, Hubbard H, Kosiak B and Dmochowski R: Staying at the cutting edge: A review and analysis of evidence reporting and grading; the recommendations of the american urological association. BJU Int 2009; **104**: 294.
6. Hsu C-C and Sandford BA: The delphi technique: Making sense of consensus. Practical assessment, research, and evaluation practical assessment, research, and evaluation 2007; **12**: 1.
7. Siegel RL, Miller KD and Jemal A: Cancer statistics, 2015. CA Cancer J Clin 2015; **65**: 5.
8. Aldousari S and Kassouf W: Update on the management of non-muscle invasive bladder cancer. Can Urol Assoc J 2010; **4**: 56.
9. Abdollah F, Gandaglia G, Thuret R et al: Incidence, survival and mortality rates of stage-specific bladder cancer in united states: A trend analysis. Cancer Epidemiol 2013; **37**: 219.
10. Nielsen ME, Smith AB, Meyer AM et al: Trends in stage-specific incidence rates for urothelial carcinoma of the bladder in the united states: 1988 to 2006. Cancer 2014; **120**: 86.
11. Freedman ND, Silverman DT, Hollenbeck AR et al: Association between smoking and risk of bladder cancer among men and women. Jama 2011; **306**: 737.
12. Rink M, Xylinas E, Babjuk M et al: Impact of smoking on outcomes of patients with a history of recurrent nonmuscle invasive bladder cancer. J Urol 2012; **188**: 2120.
13. Lammers RJ, Witjes WP, Hendricksen K et al: Smoking status is a risk factor for recurrence after transurethral resection of non-muscle-invasive bladder cancer. Eur Urol 2011; **60**: 713.
14. Kiriluk KJ, Prasad SM, Patel AR et al: Bladder cancer risk from occupational and environmental exposures. Urol Oncol 2012; **30**: 199.

15. Burger M, Catto JW, Dalbagni G et al: Epidemiology and risk factors of urothelial bladder cancer. *Eur Urol* 2013; **63**: 234.
16. Fernandes ET, Manivel JC, Reddy PK and Ercole CJ: Cyclophosphamide associated bladder cancer--a highly aggressive disease: Analysis of 12 cases. *J Urol* 1996; **156**: 1931.
17. Travis LB, Curtis RE, Boice JD, Jr. and Fraumeni JF, Jr.: Bladder cancer after chemotherapy for non-hodgkin's lymphoma. *N Engl J Med* 1989; **321**: 544.
18. van der Post RS, Kiemeney LA, Ligtenberg MJ et al: Risk of urothelial bladder cancer in lynch syndrome is increased, in particular among msh2 mutation carriers. *J Med Genet* 2010; **47**: 464.
19. Skeldon SC, Semotiuk K, Aronson M et al: Patients with lynch syndrome mismatch repair gene mutations are at higher risk for not only upper tract urothelial cancer but also bladder cancer. *Eur Urol* 2013; **63**: 379.
20. El-Bolkainy MN, Mokhtar NM, Ghoneim MA and Hussein MH: The impact of schistosomiasis on the pathology of bladder carcinoma. *Cancer* 1981; **48**: 2643.
21. Poon SL, Huang MN, Choo Y et al: Mutation signatures implicate aristolochic acid in bladder cancer development. *Genome Med* 2015; **7**: 38.
22. Bostrom PJ and Soloway MS: Secondary cancer after radiotherapy for prostate cancer: Should we be more aware of the risk? *Eur Urol* 2007; **52**: 973.
23. García-Closas M, Malats N, Silverman D et al: Nat2 slow acetylation, gstm1 null genotype, and risk of bladder cancer: Results from the spanish bladder cancer study and meta-analyses. *Lancet* 2005; **366**: 649.
24. Ploussard G, Dubosq F, Soliman H et al: Prognostic value of loss of heterozygosity at chromosome 9p in non-muscle-invasive bladder cancer. *Urology* 2010; **76**: 513.e13.
25. Krüger S, Mahnken A, Kausch I and Feller AC: P16 immunoreactivity is an independent predictor of tumor progression in minimally invasive urothelial bladder carcinoma. *Eur Urol* 2005; **47**: 463.
26. Bartoletti R, Cai T, Nesi G et al: Loss of p16 expression and chromosome 9p21 loh in predicting outcome of patients affected by superficial bladder cancer. *J Surg Res* 2007; **143**: 422.
27. Castillo-Martin M, Domingo-Domenech J, Karni-Schmidt O et al: Molecular pathways of urothelial development and bladder tumorigenesis. *Urol Oncol* 2010; **28**: 401.
28. Knowles MA and Hurst CD: Molecular biology of bladder cancer: New insights into pathogenesis and clinical diversity. *Nat Rev Cancer* 2015; **15**: 25.
29. Davis R, Jones JS, Barocas DA et al: Diagnosis, evaluation and follow-up of asymptomatic microhematuria (amh) in adults: Aua guideline. *J Urol* 2012; **188**: 2473.
30. Dimashkieh H, Wolff DJ, Smith TM et al: Evaluation of urovysion and cytology for bladder cancer detection: A study of 1835 paired urine samples with clinical and histologic correlation. *Cancer Cytopathol* 2013; **121**: 591.
31. Schroeder GL, Lorenzo-Gomez MF, Hautmann SH et al: A side by side comparison of cytology and biomarkers for bladder cancer detection. *J Urol* 2004; **172**: 1123.
32. Rhéaume-Lanoie J, Lepanto L, Fradet V et al: Diagnostic performance of ultrasound for macroscopic hematuria in the era of multidetector computed tomography urography. *Can Assoc Radiol J* 2014; **65**: 253.
33. Edge SB and Compton CC: The american joint committee on cancer: The 7th edition of the ajcc cancer staging manual and the future of tnm. *Ann Surg Oncol* 2010; **17**: 1471.
34. Lotan Y, Gupta A, Shariat SF et al: Lymphovascular invasion is independently associated with overall survival, cause-specific survival, and local and distant recurrence in patients with negative lymph nodes at radical cystectomy. *J Clin Oncol* 2005; **23**: 6533.
35. Quek ML, Stein JP, Nichols PW et al: Prognostic significance of lymphovascular invasion of bladder cancer treated with radical cystectomy. *J Urol* 2005; **174**: 103.
36. Resnick MJ, Bergey M, Magerfleisch L et al: Longitudinal evaluation of the concordance and prognostic value of lymphovascular invasion in transurethral resection and radical cystectomy specimens. *BJU Int* 2011; **107**: 46.

37. Kim HS, Kim M, Jeong CW et al: Presence of lymphovascular invasion in urothelial bladder cancer specimens after transurethral resections correlates with risk of upstaging and survival: A systematic review and meta-analysis. *Urol Oncol* 2014; **32**: 1191.
38. Eble JN, Sauter G, Epstein JI and Sesterhenn IA: World health organization classification of tumours: Pathology and genetics of tumours of the urinary and male genital organs, 2004
39. Lopez-Beltran A and Montironi R: Non-invasive urothelial neoplasms: According to the most recent who classification. *Eur Urol* 2004; **46**: 170.
40. Palou J, Sylvester RJ, Faba OR et al: Female gender and carcinoma in situ in the prostatic urethra are prognostic factors for recurrence, progression, and disease-specific mortality in t1g3 bladder cancer patients treated with bacillus calmette-guérin. *Eur Urol* 2012; **62**: 118.
41. Cookson MS, Herr HW, Zhang ZF et al: The treated natural history of high risk superficial bladder cancer: 15-year outcome. *J Urol* 1997; **158**: 62.
42. Leblanc B, Duclos AJ, Bénard F et al: Long-term followup of initial ta grade 1 transitional cell carcinoma of the bladder. *J Urol* 1999; **162**: 1946.
43. Sylvester RJ, van der Meijden AP, Oosterlinck W et al: Predicting recurrence and progression in individual patients with stage ta t1 bladder cancer using eortc risk tables: A combined analysis of 2596 patients from seven eortc trials. *Eur Urol* 2006; **49**: 466.
44. Fernandez-Gomez J, Madero R, Solsona E et al: Predicting nonmuscle invasive bladder cancer recurrence and progression in patients treated with bacillus calmette-guerin: The cueto scoring model. *J Urol* 2009; **182**: 2195.
45. Cambier S, Sylvester RJ, Collette L et al: Eortc nomograms and risk groups for predicting recurrence, progression, and disease-specific and overall survival in non-muscle-invasive stage ta-t1 urothelial bladder cancer patients treated with 1-3 years of maintenance bacillus calmette-guérin. *Eur Urol* 2016; **69**: 60.
46. Ohman EM, Granger CB, Harrington RA and Lee KL: Risk stratification and therapeutic decision making in acute coronary syndromes. *Jama* 2000; **284**: 876.
47. Fernandez-Gomez J, Madero R, Solsona E et al: The eortc tables overestimate the risk of recurrence and progression in patients with non-muscle-invasive bladder cancer treated with bacillus calmette-guérin: External validation of the eortc risk tables. *Eur Urol* 2011; **60**: 423.
48. Altieri VM, Castellucci R, Palumbo P et al: Recurrence and progression in non-muscle-invasive bladder cancer using eortc risk tables. *Urol Int* 2012; **89**: 61.
49. Rosevear HM, Lightfoot AJ, Nepple KG and O'Donnell MA: Usefulness of the spanish urological club for oncological treatment scoring model to predict nonmuscle invasive bladder cancer recurrence in patients treated with intravesical bacillus calmette-guérin plus interferon- α . *J Urol* 2011; **185**: 67.
50. Hernández V, De La Peña E, Martin MD et al: External validation and applicability of the eortc risk tables for non-muscle-invasive bladder cancer. *World J Urol* 2011; **29**: 409.
51. Vedder MM, Márquez M, de Bekker-Grob EW et al: Risk prediction scores for recurrence and progression of non-muscle invasive bladder cancer: An international validation in primary tumours. *PLoS One* 2014; **9**: e96849.
52. Xylinas E, Kent M, Kluth L et al: Accuracy of the eortc risk tables and of the cueto scoring model to predict outcomes in non-muscle-invasive urothelial carcinoma of the bladder. *Br J Cancer* 2013; **109**: 1460.
53. Shirakawa H, Kikuchi E, Tanaka N et al: Prognostic significance of bacillus calmette-guérin failure classification in non-muscle-invasive bladder cancer. *BJU Int* 2012; **110**: E216.
54. Herr HW, Milan TN and Dalbagni G: Bcg-refractory vs. Bcg-relapsing non-muscle-invasive bladder cancer: A prospective cohort outcomes study. *Urol Oncol* 2015; **33**: 108.e1.
55. Aaronson DS, Walsh TJ, Smith JF et al: Meta-analysis: Does lidocaine gel before flexible cystoscopy provide pain relief? *BJU Int* 2009; **104**: 506.
56. Smith JA, Jr.: Surgical management of superficial bladder cancer (stages ta/t1/cis). *Semin Surg Oncol* 1997; **13**: 328.

57. Brausi M, Collette L, Kurth K et al: Variability in the recurrence rate at first follow-up cystoscopy after tur in stage ta t1 transitional cell carcinoma of the bladder: A combined analysis of seven eortc studies. *Eur Urol* 2002; **41**: 523.
58. Richterstetter M, Wullich B, Amann K et al: The value of extended transurethral resection of bladder tumour (turbt) in the treatment of bladder cancer. *BJU Int* 2012; **110**: E76.
59. Herr HW and Donat SM: Quality control in transurethral resection of bladder tumours. *BJU Int* 2008; **102**: 1242.
60. Mariappan P, Zachou A and Grigor KM: Detrusor muscle in the first, apparently complete transurethral resection of bladder tumour specimen is a surrogate marker of resection quality, predicts risk of early recurrence, and is dependent on operator experience. *Eur Urol* 2010; **57**: 843.
61. Kausch I, Sommerauer M, Montorsi F et al: Photodynamic diagnosis in non-muscle-invasive bladder cancer: A systematic review and cumulative analysis of prospective studies. *Eur Urol* 2010; **57**: 595.
62. Venkatramani V, Panda A, Manojkumar R and Kekre NS: Monopolar versus bipolar transurethral resection of bladder tumors: A single center, parallel arm, randomized, controlled trial. *J Urol* 2014; **191**: 1703.
63. Sugihara T, Yasunaga H, Horiguchi H et al: Comparison of perioperative outcomes including severe bladder injury between monopolar and bipolar transurethral resection of bladder tumors: A population based comparison. *J Urol* 2014; **192**: 1355.
64. Herr HW, Donat SM and Reuter VE: Management of low grade papillary bladder tumors. *J Urol* 2007; **178**: 1201.
65. Soloway MS, Bruck DS and Kim SS: Expectant management of small, recurrent, noninvasive papillary bladder tumors. *J Urol* 2003; **170**: 438.
66. Donat SM, North A, Dalbagni G and Herr HW: Efficacy of office fulguration for recurrent low grade papillary bladder tumors less than 0.5 cm. *J Urol* 2004; **171**: 636.
67. Gofrit ON, Pode D, Lazar A et al: Watchful waiting policy in recurrent ta g1 bladder tumors. *Eur Urol* 2006; **49**: 303.
68. Palou J, Rodríguez-Rubio F, Huguet J et al: Multivariate analysis of clinical parameters of synchronous primary superficial bladder cancer and upper urinary tract tumor. *J Urol* 2005; **174**: 859.
69. Millán-Rodríguez F, Chéchile-Toniolo G, Salvador-Bayarri J et al: Upper urinary tract tumors after primary superficial bladder tumors: Prognostic factors and risk groups. *J Urol* 2000; **164**: 1183.
70. van der Meijden A, Oosterlinck W, Brausi M et al: Significance of bladder biopsies in ta,t1 bladder tumors: A report from the eortc genito-urinary tract cancer cooperative group. Eortc-gu group superficial bladder committee. *Eur Urol* 1999; **35**: 267.
71. Hara T, Takahashi M, Gondo T et al: Risk of concomitant carcinoma in situ determining biopsy candidates among primary non-muscle-invasive bladder cancer patients: Retrospective analysis of 173 japanese cases. *Int J Urol* 2009; **16**: 293.
72. Mungan MU, Canda AE, Tuzel E et al: Risk factors for mucosal prostatic urethral involvement in superficial transitional cell carcinoma of the bladder. *Eur Urol* 2005; **48**: 760.
73. Daneshmand S, Schuckman AK, Bochner BH et al: Hexaminolevulinate blue-light cystoscopy in non-muscle-invasive bladder cancer: Review of the clinical evidence and consensus statement on appropriate use in the USA. *Nat Rev Urol* 2014; **11**: 589.
74. Witjes JA, Babjuk M, Gontero P et al: Clinical and cost effectiveness of hexaminolevulinate-guided blue-light cystoscopy: Evidence review and updated expert recommendations. *Eur Urol* 2014; **66**: 863.
75. Cho KS, Seo HK, Joung JY et al: Lymphovascular invasion in transurethral resection specimens as predictor of progression and metastasis in patients with newly diagnosed t1 bladder urothelial cancer. *J Urol* 2009; **182**: 2625.
76. Huguet J, Crego M, Sabaté S et al: Cystectomy in patients with high risk superficial bladder tumors who fail intravesical bcg therapy: Pre-cystectomy prostate involvement as a prognostic factor. *Eur Urol* 2005; **48**: 53.
77. Shapur NK, Katz R, Pode D et al: Is radical cystectomy mandatory in every patient with variant histology of bladder cancer. *Rare Tumors* 2011; **3**: e22.

78. Hansel DE, Amin MB, Comperat E et al: A contemporary update on pathology standards for bladder cancer: Transurethral resection and radical cystectomy specimens. *European Urology* 2013; **63**: 321.
79. Linder BJ, Boorjian SA, Cheville JC et al: The impact of histological reclassification during pathology re-review--evidence of a will rogers effect in bladder cancer? *J Urol* 2013; **190**: 1692.
80. Kim SP, Frank I, Cheville JC et al: The impact of squamous and glandular differentiation on survival after radical cystectomy for urothelial carcinoma. *J Urol* 2012; **188**: 405.
81. Linder BJ, Frank I, Cheville JC et al: Outcomes following radical cystectomy for nested variant of urothelial carcinoma: A matched cohort analysis. *J Urol* 2013; **189**: 1670.
82. Wang JK, Boorjian SA, Cheville JC et al: Outcomes following radical cystectomy for micropapillary bladder cancer versus pure urothelial carcinoma: A matched cohort analysis. *World J Urol* 2012; **30**: 801.
83. Wasco MJ, Daignault S, Zhang Y et al: Urothelial carcinoma with divergent histologic differentiation (mixed histologic features) predicts the presence of locally advanced bladder cancer when detected at transurethral resection. *Urology* 2007; **70**: 69.
84. Iida K, Miyake M, Murakami K et al: Bacillus calmette-guérin-unresponsive non-muscle invasive bladder cancer outcomes in patients without radical cystectomy. *Int J Clin Oncol* 2021; **26**: 2104.
85. Daneshmand S: Determining the role of cystectomy for high-grade t1 urothelial carcinoma. *Urol Clin North Am* 2013; **40**: 233.
86. Kamat AM, Gee JR, Dinney CP et al: The case for early cystectomy in the treatment of nonmuscle invasive micropapillary bladder carcinoma. *J Urol* 2006; **175**: 881.
87. Papanicolaou GN and Marshall VF: Urine sediment smears as a diagnostic procedure in cancers of the urinary tract. *Science* 1945; **101**: 519.
88. Karakiewicz PI, Benayoun S, Zippe C et al: Institutional variability in the accuracy of urinary cytology for predicting recurrence of transitional cell carcinoma of the bladder. *BJU Int* 2006; **97**: 997.
89. Tomasini JM and Konety BR: Urinary markers/cytology: What and when should a urologist use. *Urol Clin North Am* 2013; **40**: 165.
90. O'Sullivan P, Sharples K, Dalphin M et al: A multigene urine test for the detection and stratification of bladder cancer in patients presenting with hematuria. *J Urol* 2012; **188**: 741.
91. Chou R, Buckley D, Fu R et al: Ahrq comparative effectiveness reviews. In: Emerging approaches to diagnosis and treatment of non-muscle-invasive bladder cancer. Rockville (MD): Agency for Healthcare Research and Quality (US), 2015
92. Poulakis V, Witzsch U, De Vries R et al: A comparison of urinary nuclear matrix protein-22 and bladder tumour antigen tests with voided urinary cytology in detecting and following bladder cancer: The prognostic value of false-positive results. *BJU Int* 2001; **88**: 692.
93. Kavalieris L, O'Sullivan P, Frampton C et al: Performance characteristics of a multigene urine biomarker test for monitoring for recurrent urothelial carcinoma in a multicenter study. *J Urol* 2017; **197**: 1419.
94. Valenberg F, Hiar AM, Wallace E et al: Prospective validation of an mrna-based urine test for surveillance of patients with bladder cancer. *Eur Urol* 2019; **75**: 853.
95. Witjes JA, Morote J, Cornel EB et al: Performance of the bladder epicheck™ methylation test for patients under surveillance for non-muscle-invasive bladder cancer: Results of a multicenter, prospective, blinded clinical trial. *Eur Urol Oncol* 2018; **1**: 307.
96. Lotan Y, O'Sullivan P, Raman JD et al: Clinical comparison of noninvasive urine tests for ruling out recurrent urothelial carcinoma. *Urol Oncol* 2017; **35**: 531.e15.
97. Steiner H, Bergmeister M, Verdorfer I et al: Early results of bladder-cancer screening in a high-risk population of heavy smokers. *BJU Int* 2008; **102**: 291.
98. Lotan Y, Elias K, Svatek RS et al: Bladder cancer screening in a high risk asymptomatic population using a point of care urine based protein tumor marker. *J Urol* 2009; **182**: 52.

99. Bangma CH, Loeb S, Busstra M et al: Outcomes of a bladder cancer screening program using home hematuria testing and molecular markers. *Eur Urol* 2013; **64**: 41.
100. Huber S, Schwentner C, Taeger D et al: Nuclear matrix protein-22: A prospective evaluation in a population at risk for bladder cancer. Results from the uroscreen study. *BJU Int* 2012; **110**: 699.
101. van Rhijn BW, van der Poel HG and van der Kwast TH: Urine markers for bladder cancer surveillance: A systematic review. *Eur Urol* 2005; **47**: 736.
102. Lotan Y and Roehrborn CG: Sensitivity and specificity of commonly available bladder tumor markers versus cytology: Results of a comprehensive literature review and meta-analyses. *Urology* 2003; **61**: 109.
103. van der Aa MN, Steyerberg EW, Bangma C et al: Cystoscopy revisited as the gold standard for detecting bladder cancer recurrence: Diagnostic review bias in the randomized, prospective cefub trial. *J Urol* 2010; **183**: 76.
104. Kamat AM, Dickstein RJ, Messetti F et al: Use of fluorescence in situ hybridization to predict response to bacillus calmette-guérin therapy for bladder cancer: Results of a prospective trial. *J Urol* 2012; **187**: 862.
105. Mengual L, Marín-Aguilera M, Ribal MJ et al: Clinical utility of fluorescent in situ hybridization for the surveillance of bladder cancer patients treated with bacillus calmette-guérin therapy. *Eur Urol* 2007; **52**: 752.
106. Whitson J, Berry A, Carroll P and Konety B: A multicolour fluorescence in situ hybridization test predicts recurrence in patients with high-risk superficial bladder tumours undergoing intravesical therapy. *BJU Int* 2009; **104**: 336.
107. Kipp BR, Karnes RJ, Brankley SM et al: Monitoring intravesical therapy for superficial bladder cancer using fluorescence in situ hybridization. *J Urol* 2005; **173**: 401.
108. Savic S, Zlobec I, Thalmann GN et al: The prognostic value of cytology and fluorescence in situ hybridization in the follow-up of nonmuscle-invasive bladder cancer after intravesical bacillus calmette-guérin therapy. *Int J Cancer* 2009; **124**: 2899.
109. Shefer HK, Masarwe I, Bejar J et al: Performance of celldetect for detection of bladder cancer: Comparison with urine cytology and urovysion. *Urol Oncol* 2023; **41**: 296.e1.
110. Mokhtar GA, Al-Dousari M and Al-Ghamedi D: Diagnostic significance of atypical category in the voided urine samples: A retrospective study in a tertiary care center. *Urol Ann* 2010; **2**: 100.
111. Nabi G, Greene D and O'Donnell MO: Suspicious urinary cytology with negative evaluation for malignancy in the diagnostic investigation of haematuria: How to follow up? *J Clin Pathol* 2004; **57**: 365.
112. Raitanen MP, Aine R, Kylmä T et al: The dilemma of suspicious urine cytology in patients being followed for bladder cancer. *Ann Chir Gynaecol* 2001; **90**: 256.
113. Cochetti G, Rossi de Vermandois JA, Maulà V et al: Diagnostic performance of the bladder epicheck methylation test and photodynamic diagnosis-guided cystoscopy in the surveillance of high-risk non-muscle invasive bladder cancer: A single centre, prospective, blinded clinical trial. *Urol Oncol* 2022; **40**: 105.e11.
114. Pierconti F, Martini M, Cenci T et al: The bladder epicheck test and cytology in the follow-up of patients with non-muscle-invasive high grade bladder carcinoma. *Urol Oncol* 2022; **40**: 108.e19.
115. Trenti E, Pycha S, Mian C et al: Comparison of 2 new real-time polymerase chain reaction-based urinary markers in the follow-up of patients with non-muscle-invasive bladder cancer. *Cancer Cytopathol* 2020; **128**: 341.
116. Sarosdy MF, Schellhammer P, Bokinsky G et al: Clinical evaluation of a multi-target fluorescent in situ hybridization assay for detection of bladder cancer. *J Urol* 2002; **168**: 1950.
117. Yoder BJ, Skacel M, Hedgepeth R et al: Reflex urovysion testing of bladder cancer surveillance patients with equivocal or negative urine cytology: A prospective study with focus on the natural history of anticipatory positive findings. *Am J Clin Pathol* 2007; **127**: 295.
118. Seideman C, Canter D, Kim P et al: Multicenter evaluation of the role of urovysion fish assay in surveillance of patients with bladder cancer: Does fish positivity anticipate recurrence? *World J Urol* 2015; **33**: 1309.
119. Filbeck T, Roessler W, Knuechel R et al: 5-aminolevulinic acid-induced fluorescence endoscopy applied at secondary transurethral resection after conventional resection of primary superficial bladder tumors. *Urology* 1999; **53**: 77.

120. Draga RO, Grimbergen MC, Kok ET et al: Predictors of false positives in 5-aminolevulinic acid-induced photodynamic diagnosis of bladder carcinoma: Identification of patient groups that may benefit most from highly specific optical diagnostics. *Urology* 2009; **74**: 851.
121. Witjes JA, Redorta JP, Jacqmin D et al: Hexaminolevulinate-guided fluorescence cystoscopy in the diagnosis and follow-up of patients with non-muscle-invasive bladder cancer: Review of the evidence and recommendations. *Eur Urol* 2010; **57**: 607.
122. Herr HW: Role of repeat resection in non-muscle-invasive bladder cancer. *J Natl Compr Canc Netw* 2015; **13**: 1041.
123. Vianello A, Costantini E, Del Zingaro M et al: Repeated white light transurethral resection of the bladder in nonmuscle-invasive urothelial bladder cancers: Systematic review and meta-analysis. *J Endourol* 2011; **25**: 1703.
124. Grimm MO, Steinhoff C, Simon X et al: Effect of routine repeat transurethral resection for superficial bladder cancer: A long-term observational study. *J Urol* 2003; **170**: 433.
125. Herr HW: The value of a second transurethral resection in evaluating patients with bladder tumors. *J Urol* 1999; **162**: 74.
126. Divrik RT, Yildirim U, Zorlu F and Ozen H: The effect of repeat transurethral resection on recurrence and progression rates in patients with t1 tumors of the bladder who received intravesical mitomycin: A prospective, randomized clinical trial. *J Urol* 2006; **175**: 1641.
127. Herr HW, Donat SM and Dalbagni G: Can restaging transurethral resection of t1 bladder cancer select patients for immediate cystectomy? *J Urol* 2007; **177**: 75.
128. Sfakianos JP, Kim PH, Hakimi AA and Herr HW: The effect of restaging transurethral resection on recurrence and progression rates in patients with nonmuscle invasive bladder cancer treated with intravesical bacillus calmette-guérin. *J Urol* 2014; **191**: 341.
129. Guevara A, Salomon L, Allory Y et al: The role of tumor-free status in repeat resection before intravesical bacillus calmette-guerin for high grade ta, t1 and cis bladder cancer. *J Urol* 2010; **183**: 2161.
130. Herr HW: Restaging transurethral resection of high risk superficial bladder cancer improves the initial response to bacillus calmette-guerin therapy. *J Urol* 2005; **174**: 2134.
131. Divrik RT, Sahin AF, Yildirim U et al: Impact of routine second transurethral resection on the long-term outcome of patients with newly diagnosed pt1 urothelial carcinoma with respect to recurrence, progression rate, and disease-specific survival: A prospective randomised clinical trial. *Eur Urol* 2010; **58**: 185.
132. Pan JS, Slocum HK, Rustum YM et al: Inhibition of implantation of murine bladder tumor by thiotepa in cauterized bladder. *J Urol* 1989; **142**: 1589.
133. Brocks CP, Büttner H and Böhle A: Inhibition of tumor implantation by intravesical gemcitabine in a murine model of superficial bladder cancer. *J Urol* 2005; **174**: 1115.
134. Oosterlinck W, Kurth KH, Schröder F et al: A prospective european organization for research and treatment of cancer genitourinary group randomized trial comparing transurethral resection followed by a single intravesical instillation of epirubicin or water in single stage ta, t1 papillary carcinoma of the bladder. *J Urol* 1993; **149**: 749.
135. Taylor CA, Tobert CM, Kahnoski RJ et al: The value of a comprehensive primary outcome - results of a negative randomized control trial in the non-muscle invasive bladder cancer population. *Can J Urol* 2021; **28**: 10756.
136. Sylvester RJ, Oosterlinck W and van der Meijden AP: A single immediate postoperative instillation of chemotherapy decreases the risk of recurrence in patients with stage ta t1 bladder cancer: A meta-analysis of published results of randomized clinical trials. *J Urol* 2004; **171**: 2186.
137. Abern MR, Owusu RA, Anderson MR et al: Perioperative intravesical chemotherapy in non-muscle-invasive bladder cancer: A systematic review and meta-analysis. *J Natl Compr Canc Netw* 2013; **11**: 477.
138. Perlis N, Zlotta AR, Beyene J et al: Immediate post-transurethral resection of bladder tumor intravesical chemotherapy prevents non-muscle-invasive bladder cancer recurrences: An updated meta-analysis on 2548 patients and quality-of-evidence review. *Eur Urol* 2013; **64**: 421.
139. Kang M, Jeong CW, Kwak C et al: Single, immediate postoperative instillation of chemotherapy in non-muscle invasive bladder cancer: A systematic review and network meta-analysis of randomized clinical trials using different drugs.

Oncotarget 2016; **7**: 45479.

140. Daryanto B, Purnomo AF, Seputra KP and Budaya TN: Comparison between intravesical chemotherapy epirubicin and mitomycin-c after turb vs turb alone with recurrence rate of non-muscle invasive bladder cancer: Meta-analysis. Med Arch 2022; **76**: 198.
141. Berrum-Svennung I, Granfors T, Jahnson S et al: A single instillation of epirubicin after transurethral resection of bladder tumors prevents only small recurrences. J Urol 2008; **179**: 101.
142. Gudjónsson S, Adell L, Merdasa F et al: Should all patients with non-muscle-invasive bladder cancer receive early intravesical chemotherapy after transurethral resection? The results of a prospective randomised multicentre study. Eur Urol 2009; **55**: 773.
143. Boufflioux C, Kurth KH, Bono A et al: Intravesical adjuvant chemotherapy for superficial transitional cell bladder carcinoma: Results of 2 european organization for research and treatment of cancer randomized trials with mitomycin c and doxorubicin comparing early versus delayed instillations and short-term versus long-term treatment. European organization for research and treatment of cancer genitourinary group. J Urol 1995; **153**: 934.
144. Kaasinen E, Rintala E, Hellström P et al: Factors explaining recurrence in patients undergoing chemoimmunotherapy regimens for frequently recurring superficial bladder carcinoma. Eur Urol 2002; **42**: 167.
145. Sylvester RJ, Oosterlinck W and Witjes JA: The schedule and duration of intravesical chemotherapy in patients with non-muscle-invasive bladder cancer: A systematic review of the published results of randomized clinical trials. Eur Urol 2008; **53**: 709.
146. Sylvester RJ, Oosterlinck W, Holmang S et al: Systematic review and individual patient data meta-analysis of randomized trials comparing a single immediate instillation of chemotherapy after transurethral resection with transurethral resection alone in patients with stage pta-pt1 urothelial carcinoma of the bladder: Which patients benefit from the instillation? Eur Urol 2016; **69**: 231.
147. Pode D, Alon Y, Horowitz AT et al: The mechanism of human bladder tumor implantation in an in vitro model. J Urol 1986; **136**: 482.
148. Günther JH, Jurczok A, Wulf T et al: Optimizing syngeneic orthotopic murine bladder cancer (mb49). Cancer Res 1999; **59**: 2834.
149. Böhle A, Jurczok A, Ardelt P et al: Inhibition of bladder carcinoma cell adhesion by oligopeptide combinations in vitro and in vivo. J Urol 2002; **167**: 357.
150. Oddens JR, van der Meijden AP and Sylvester R: One immediate postoperative instillation of chemotherapy in low risk ta, t1 bladder cancer patients. Is it always safe? Eur Urol 2004; **46**: 336.
151. Elmamoun MH, Christmas TJ and Woodhouse CR: Destruction of the bladder by single dose mitomycin c for low-stage transitional cell carcinoma (tcc)--avoidance, recognition, management and consent. BJU Int 2014; **113**: E34.
152. Tolley DA, Parmar MK, Grigor KM et al: The effect of intravesical mitomycin c on recurrence of newly diagnosed superficial bladder cancer: A further report with 7 years of follow up. J Urol 1996; **155**: 1233.
153. Ali-el-Dein B, el-Baz M, Aly AN et al: Intravesical epirubicin versus doxorubicin for superficial bladder tumors (stages pta and pt1): A randomized prospective study. J Urol 1997; **158**: 68.
154. Liu B, Wang Z, Chen B et al: Randomized study of single instillation of epirubicin for superficial bladder carcinoma: Long-term clinical outcomes. Cancer Invest 2006; **24**: 160.
155. Türkeri L, Tanıdır Y, Çal Ç et al: Comparison of the efficacy of single or double intravesical epirubicin instillation in the early postoperative period to prevent recurrences in non-muscle-invasive urothelial carcinoma of the bladder: Prospective, randomized multicenter study. Urol Int 2010; **85**: 261.
156. Au JL, Badalament RA, Wientjes MG et al: Methods to improve efficacy of intravesical mitomycin c: Results of a randomized phase iii trial. J Natl Cancer Inst 2001; **93**: 597.
157. Böhle A, Jocham D and Bock PR: Intravesical bacillus calmette-guerin versus mitomycin c for superficial bladder cancer: A formal meta-analysis of comparative studies on recurrence and toxicity. J Urol 2003; **169**: 90.
158. Han RF and Pan JG: Can intravesical bacillus calmette-guérin reduce recurrence in patients with superficial bladder cancer? A meta-analysis of randomized trials. Urology 2006; **67**: 1216.

159. Shelley MD, Kynaston H, Court J et al: A systematic review of intravesical bacillus calmette-guérin plus transurethral resection vs transurethral resection alone in ta and t1 bladder cancer. *BJU Int* 2001; **88**: 209.
160. Shelley MD, Wilt TJ, Court J et al: Intravesical bacillus calmette-guérin is superior to mitomycin c in reducing tumour recurrence in high-risk superficial bladder cancer: A meta-analysis of randomized trials. *BJU Int* 2004; **93**: 485.
161. Herr HW and Morales A: History of bacillus calmette-guerin and bladder cancer: An immunotherapy success story. *J Urol* 2008; **179**: 53.
162. Rentsch CA, Birkhäuser FD, Biot C et al: Bacillus calmette-guérin strain differences have an impact on clinical outcome in bladder cancer immunotherapy. *Eur Urol* 2014; **66**: 677.
163. Zhu S, Tang Y, Li K et al: Optimal schedule of bacillus calmette-guerin for non-muscle-invasive bladder cancer: A meta-analysis of comparative studies. *BMC Cancer* 2013; **13**: 332.
164. Oddens J, Brausi M, Sylvester R et al: Final results of an eortc-gu cancers group randomized study of maintenance bacillus calmette-guérin in intermediate- and high-risk ta, t1 papillary carcinoma of the urinary bladder: One-third dose versus full dose and 1 year versus 3 years of maintenance. *Eur Urol* 2013; **63**: 462.
165. Houghton BB, Chalasani V, Hayne D et al: Intravesical chemotherapy plus bacille calmette-guérin in non-muscle invasive bladder cancer: A systematic review with meta-analysis. *BJU Int* 2013; **111**: 977.
166. Solsona E, Madero R, Chantada V et al: Sequential combination of mitomycin c plus bacillus calmette-guérin (bcg) is more effective but more toxic than bcg alone in patients with non-muscle-invasive bladder cancer in intermediate- and high-risk patients: Final outcome of cueto 93009, a randomized prospective trial. *Eur Urol* 2015; **67**: 508.
167. Nepple KG, Lightfoot AJ, Rosevear HM et al: Bacillus calmette-guérin with or without interferon α -2b and megadose versus recommended daily allowance vitamins during induction and maintenance intravesical treatment of nonmuscle invasive bladder cancer. *J Urol* 2010; **184**: 1915.
168. Arends TJ, Nativ O, Maffezzini M et al: Results of a randomised controlled trial comparing intravesical chemohyperthermia with mitomycin c versus bacillus calmette-guérin for adjuvant treatment of patients with intermediate- and high-risk non-muscle-invasive bladder cancer. *Eur Urol* 2016; **69**: 1046.
169. Friedrich MG, Pichlmeier U, Schwaibold H et al: Long-term intravesical adjuvant chemotherapy further reduces recurrence rate compared with short-term intravesical chemotherapy and short-term therapy with bacillus calmette-guérin (bcg) in patients with non-muscle-invasive bladder carcinoma. *Eur Urol* 2007; **52**: 1123.
170. Okamura K, Kinukawa T, Tsumura Y et al: A randomized study of short-versus long-term intravesical epirubicin instillation for superficial bladder cancer. Nagoya university urological oncology group. *Eur Urol* 1998; **33**: 285.
171. Serretta V, Morgia G, Altieri V et al: A 1-year maintenance after early adjuvant intravesical chemotherapy has a limited efficacy in preventing recurrence of intermediate risk non-muscle-invasive bladder cancer. *BJU Int* 2010; **106**: 212.
172. Hendricksen K, Witjes WP, Idema JG et al: Comparison of three schedules of intravesical epirubicin in patients with non-muscle-invasive bladder cancer. *Eur Urol* 2008; **53**: 984.
173. Koga H, Kuroiwa K, Yamaguchi A et al: A randomized controlled trial of short-term versus long-term prophylactic intravesical instillation chemotherapy for recurrence after transurethral resection of ta/t1 transitional cell carcinoma of the bladder. *J Urol* 2004; **171**: 153.
174. Kuroda M, Niijima T, Kotake T et al: Effect of prophylactic treatment with intravesical epirubicin on recurrence of superficial bladder cancer--the 6th trial of the japanese urological cancer research group (jucrg): A randomized trial of intravesical epirubicin at dose of 20mg/40ml, 30mg/40ml, 40mg/40ml. *Eur Urol* 2004; **45**: 600.
175. Mitsumori K, Tsuchiya N, Habuchi T et al: Early and large-dose intravesical instillation of epirubicin to prevent superficial bladder carcinoma recurrence after transurethral resection. *BJU Int* 2004; **94**: 317.
176. Nomata K, Noguchi M, Kanetake H et al: Intravesical adjuvant chemotherapy for superficial transitional cell bladder carcinoma: Results of a randomized trial with epirubicin comparing short-term versus long-term maintenance treatment. *Cancer Chemother Pharmacol* 2002; **50**: 266.
177. Rübber H, Lutzeyer W, Fischer N et al: Natural history and treatment of low and high risk superficial bladder tumors. *J Urol* 1988; **139**: 283.

178. Flamm J: Long-term versus short-term doxorubicin hydrochloride instillation after transurethral resection of superficial bladder cancer. *Eur Urol* 1990; **17**: 119.
179. Brausi M, Oddens J, Sylvester R et al: Side effects of bacillus calmette-guérin (bcg) in the treatment of intermediate- and high-risk ta, t1 papillary carcinoma of the bladder: Results of the eortc genito-urinary cancers group randomised phase 3 study comparing one-third dose with full dose and 1 year with 3 years of maintenance bcg. *Eur Urol* 2014; **65**: 69.
180. Ehdaie B, Sylvester R and Herr HW: Maintenance bacillus calmette-guérin treatment of non-muscle-invasive bladder cancer: A critical evaluation of the evidence. *Eur Urol* 2013; **64**: 579.
181. Malmström PU, Sylvester RJ, Crawford DE et al: An individual patient data meta-analysis of the long-term outcome of randomised studies comparing intravesical mitomycin c versus bacillus calmette-guérin for non-muscle-invasive bladder cancer. *Eur Urol* 2009; **56**: 247.
182. Sylvester RJ, van der MA and Lamm DL: Intravesical bacillus calmette-guerin reduces the risk of progression in patients with superficial bladder cancer: A meta-analysis of the published results of randomized clinical trials. *J Urol* 2002; **168**: 1964.
183. Lamm DL, Blumenstein BA, Crissman JD et al: Maintenance bacillus calmette-guerin immunotherapy for recurrent ta, t1 and carcinoma in situ transitional cell carcinoma of the bladder: A randomized southwest oncology group study. *J Urol* 2000; **163**: 1124.
184. Herr HW and Donat SM: Prostatic tumor relapse in patients with superficial bladder tumors: 15-year outcome. *J Urol* 1999; **161**: 1854.
185. Herr HW: Extravesical tumor relapse in patients with superficial bladder tumors. *J Clin Oncol* 1998; **16**: 1099.
186. Schwalb DM, Herr HW and Fair WR: The management of clinically unconfirmed positive urinary cytology. *J Urol* 1993; **150**: 1751.
187. Canales BK, Anderson JK, Premoli J and Slaton JW: Risk factors for upper tract recurrence in patients undergoing long-term surveillance for stage ta bladder cancer. *J Urol* 2006; **175**: 74.
188. Giannarini G, Birkhäuser FD, Recker F et al: Bacillus calmette-guérin failure in patients with non-muscle-invasive urothelial carcinoma of the bladder may be due to the urologist's failure to detect urothelial carcinoma of the upper urinary tract and urethra. *Eur Urol* 2014; **65**: 825.
189. Wright JL, Hotaling J and Porter MP: Predictors of upper tract urothelial cell carcinoma after primary bladder cancer: A population based analysis. *J Urol* 2009; **181**: 1035.
190. Hurle R, Losa A, Manzetti A and Lembo A: Upper urinary tract tumors developing after treatment of superficial bladder cancer: 7-year follow-up of 591 consecutive patients. *Urology* 1999; **53**: 1144.
191. Picozzi S, Ricci C, Gaeta M et al: Upper urinary tract recurrence following radical cystectomy for bladder cancer: A meta-analysis on 13,185 patients. *J Urol* 2012; **188**: 2046.
192. Sakamoto N, Tsuneyoshi M, Naito S and Kumazawa J: An adequate sampling of the prostate to identify prostatic involvement by urothelial carcinoma in bladder cancer patients. *J Urol* 1993; **149**: 318.
193. Aharony S, Baniel J and Yossepowitch O: Clinically unconfirmed positive urinary cytology: Diagnostic implications and oncological outcomes. *BJU Int* 2011; **108**: E179.
194. Rink M, Babjuk M, Catto JW et al: Hexyl aminolevulinate-guided fluorescence cystoscopy in the diagnosis and follow-up of patients with non-muscle-invasive bladder cancer: A critical review of the current literature. *Eur Urol* 2013; **64**: 624.
195. Karl A, Tritschler S, Stanislaus P et al: Positive urine cytology but negative white-light cystoscopy: An indication for fluorescence cystoscopy? *BJU Int* 2009; **103**: 484.
196. Ray ER, Chatterton K, Khan MS et al: Hexylaminolaevulinate 'blue light' fluorescence cystoscopy in the investigation of clinically unconfirmed positive urine cytology. *BJU Int* 2009; **103**: 1363.
197. Ray ER, Chatterton K, Khan MS et al: Hexylaminolaevulinate fluorescence cystoscopy in patients previously treated with intravesical bacille calmette-guérin. *BJU Int* 2010; **105**: 789.

198. Bui TT and Schellhammer PF: Additional bacillus calmette-guérin therapy for recurrent transitional cell carcinoma after an initial complete response. *Urology* 1997; **49**: 687.
199. Brake M, Loertzer H, Horsch R and Keller H: Long-term results of intravesical bacillus calmette-guérin therapy for stage t1 superficial bladder cancer. *Urology* 2000; **55**: 673.
200. Catalona WJ, Hudson MA, Gillen DP et al: Risks and benefits of repeated courses of intravesical bacillus calmette-guerin therapy for superficial bladder cancer. *J Urol* 1987; **137**: 220.
201. de Reijke TM, Kurth KH, Sylvester RJ et al: Bacillus calmette-guerin versus epirubicin for primary, secondary or concurrent carcinoma in situ of the bladder: Results of a european organization for the research and treatment of cancer--genito-urinary group phase iii trial (30906). *J Urol* 2005; **173**: 405.
202. Babjuk M, Burger M, Zigeuner R et al: Eau guidelines on non-muscle-invasive urothelial carcinoma of the bladder: Update 2013. *Eur Urol* 2013; **64**: 639.
203. Sylvester RJ, van der Meijden A, Witjes JA et al: High-grade ta urothelial carcinoma and carcinoma in situ of the bladder. *Urology* 2005; **66**: 90.
204. Joudi FN, Smith BJ and O'Donnell MA: Final results from a national multicenter phase ii trial of combination bacillus calmette-guérin plus interferon alpha-2b for reducing recurrence of superficial bladder cancer. *Urol Oncol* 2006; **24**: 344.
205. Lam JS, Benson MC, O'Donnell MA et al: Bacillus calmete-guérin plus interferon-alpha2b intravesical therapy maintains an extended treatment plan for superficial bladder cancer with minimal toxicity. *Urol Oncol* 2003; **21**: 354.
206. O'Donnell MA, Krohn J and DeWolf WC: Salvage intravesical therapy with interferon-alpha 2b plus low dose bacillus calmette-guerin is effective in patients with superficial bladder cancer in whom bacillus calmette-guerin alone previously failed. *J Urol* 2001; **166**: 1300.
207. Gallagher BL, Joudi FN, Maymí JL and O'Donnell MA: Impact of previous bacille calmette-guérin failure pattern on subsequent response to bacille calmette-guérin plus interferon intravesical therapy. *Urology* 2008; **71**: 297.
208. Di Lorenzo G, Perdonà S, Damiano R et al: Gemcitabine versus bacille calmette-guérin after initial bacille calmette-guérin failure in non-muscle-invasive bladder cancer: A multicenter prospective randomized trial. *Cancer* 2010; **116**: 1893.
209. Solsona E, Iborra I, Dumont R et al: The 3-month clinical response to intravesical therapy as a predictive factor for progression in patients with high risk superficial bladder cancer. *J Urol* 2000; **164**: 685.
210. Herr HW and Sogani PC: Does early cystectomy improve the survival of patients with high risk superficial bladder tumors? *J Urol* 2001; **166**: 1296.
211. Schrier BP, Hollander MP, van Rhijn BW et al: Prognosis of muscle-invasive bladder cancer: Difference between primary and progressive tumours and implications for therapy. *Eur Urol* 2004; **45**: 292.
212. van den Bosch S and Alfred Witjes J: Long-term cancer-specific survival in patients with high-risk, non-muscle-invasive bladder cancer and tumour progression: A systematic review. *Eur Urol* 2011; **60**: 493.
213. Breau RH, Karnes RJ, Farmer SA et al: Progression to detrusor muscle invasion during urothelial carcinoma surveillance is associated with poor prognosis. *BJU Int* 2014; **113**: 900.
214. Raj GV, Herr H, Serio AM et al: Treatment paradigm shift may improve survival of patients with high risk superficial bladder cancer. *J Urol* 2007; **177**: 1283.
215. Yates DR, Brausi MA, Catto JW et al: Treatment options available for bacillus calmette-guérin failure in non-muscle-invasive bladder cancer. *Eur Urol* 2012; **62**: 1088.
216. Burger M, Oosterlinck W, Konety B et al: Icdud-eau international consultation on bladder cancer 2012: Non-muscle-invasive urothelial carcinoma of the bladder. *Eur Urol* 2013; **63**: 36.
217. Kamat AM, Sylvester RJ, Böhle A et al: Definitions, end points, and clinical trial designs for non-muscle-invasive bladder cancer: Recommendations from the international bladder cancer group. *J Clin Oncol* 2016; **34**: 1935.
218. Jarow JP, Lerner SP, Kluetz PG et al: Clinical trial design for the development of new therapies for nonmuscle-invasive bladder cancer: Report of a food and drug administration and american urological association public workshop.

Urology 2014; **83**: 262.

219. Lerner SP, Dinney C, Kamat A et al: Clarification of bladder cancer disease states following treatment of patients with intravesical bcg. *Bladder Cancer* 2015; **1**: 29.
220. Dinney CP, Greenberg RE and Steinberg GD: Intravesical valrubicin in patients with bladder carcinoma in situ and contraindication to or failure after bacillus calmette-guérin. *Urol Oncol* 2013; **31**: 1635.
221. Lee A: Nadofaragene firadenovec: First approval. *Drugs* 2023; **83**: 353.
222. Boorjian SA, Alemozaffar M, Konety BR et al: Intravesical nadofaragene firadenovec gene therapy for bcg-unresponsive non-muscle-invasive bladder cancer: A single-arm, open-label, repeat-dose clinical trial. *Lancet Oncol* 2021; **22**: 107.
223. Dalbagni G, Russo P, Bochner B et al: Phase ii trial of intravesical gemcitabine in bacille calmette-guérin-refractory transitional cell carcinoma of the bladder. *J Clin Oncol* 2006; **24**: 2729.
224. Addeo R, Caraglia M, Bellini S et al: Randomized phase iii trial on gemcitabine versus mytomicin in recurrent superficial bladder cancer: Evaluation of efficacy and tolerance. *J Clin Oncol* 2010; **28**: 543.
225. Cockerill PA, Knoedler JJ, Frank I et al: Intravesical gemcitabine in combination with mitomycin c as salvage treatment in recurrent non-muscle-invasive bladder cancer. *BJU Int* 2016; **117**: 456.
226. McKiernan JM, Masson P, Murphy AM et al: Phase i trial of intravesical docetaxel in the management of superficial bladder cancer refractory to standard intravesical therapy. *J Clin Oncol* 2006; **24**: 3075.
227. Barlow LJ, McKiernan JM and Benson MC: The novel use of intravesical docetaxel for the treatment of non-muscle invasive bladder cancer refractory to bcg therapy: A single institution experience. *World J Urol* 2009; **27**: 331.
228. Steinberg RL, Thomas LJ, Brooks N et al: Multi-institution evaluation of sequential gemcitabine and docetaxel as rescue therapy for nonmuscle invasive bladder cancer. *J Urol* 2020; **203**: 902.
229. McKiernan JM, Barlow LJ, Laudano MA et al: A phase i trial of intravesical nanoparticle albumin-bound paclitaxel in the treatment of bacillus calmette-guérin refractory nonmuscle invasive bladder cancer. *J Urol* 2011; **186**: 448.
230. Balar AV, Kamat AM, Kulkarni GS et al: Pembrolizumab monotherapy for the treatment of high-risk non-muscle-invasive bladder cancer unresponsive to bcg (keynote-057): An open-label, single-arm, multicentre, phase 2 study. *Lancet Oncol* 2021; **22**: 919.
231. Huncharek M, McGarry R and Kupelnick B: Impact of intravesical chemotherapy on recurrence rate of recurrent superficial transitional cell carcinoma of the bladder: Results of a meta-analysis. *Anticancer Res* 2001; **21**: 765.
232. Sylvester RJ, Brausi MA, Kirkels WJ et al: Long-term efficacy results of eortc genito-urinary group randomized phase 3 study 30911 comparing intravesical instillations of epirubicin, bacillus calmette-guérin, and bacillus calmette-guérin plus isoniazid in patients with intermediate- and high-risk stage ta t1 urothelial carcinoma of the bladder. *Eur Urol* 2010; **57**: 766.
233. Shang PF, Kwong J, Wang ZP et al: Intravesical bacillus calmette-guérin versus epirubicin for ta and t1 bladder cancer. *Cochrane Database Syst Rev* 2011: Cd006885.
234. Fritsche HM, Burger M, Svatek RS et al: Characteristics and outcomes of patients with clinical t1 grade 3 urothelial carcinoma treated with radical cystectomy: Results from an international cohort. *Eur Urol* 2010; **57**: 300.
235. Turker P, Bostrom PJ, Wroclawski ML et al: Upstaging of urothelial cancer at the time of radical cystectomy: Factors associated with upstaging and its effect on outcome. *BJU Int* 2012; **110**: 804.
236. Chalasani V, Kassouf W, Chin JL et al: Radical cystectomy for the treatment of t1 bladder cancer: The canadian bladder cancer network experience. *Can Urol Assoc J* 2011; **5**: 83.
237. Shariat SF, Palapattu GS, Karakiewicz PI et al: Discrepancy between clinical and pathologic stage: Impact on prognosis after radical cystectomy. *Eur Urol* 2007; **51**: 137.
238. Parmar MK, Freedman LS, Hargreave TB and Tolley DA: Prognostic factors for recurrence and followup policies in the treatment of superficial bladder cancer: Report from the british medical research council subgroup on superficial bladder cancer (urological cancer working party). *J Urol* 1989; **142**: 284.

239. Kurth KH, Denis L, Bouffieux C et al: Factors affecting recurrence and progression in superficial bladder tumours. *Eur J Cancer* 1995; **31a**: 1840.
240. Martin-Doyle W, Leow JJ, Orsola A et al: Improving selection criteria for early cystectomy in high-grade t1 bladder cancer: A meta-analysis of 15,215 patients. *J Clin Oncol* 2015; **33**: 643.
241. Bianco FJ, Jr., Justa D, Grignon DJ et al: Management of clinical t1 bladder transitional cell carcinoma by radical cystectomy. *Urol Oncol* 2004; **22**: 290.
242. Stein JP, Lieskovsky G, Cote R et al: Radical cystectomy in the treatment of invasive bladder cancer: Long-term results in 1,054 patients. *J Clin Oncol* 2001; **19**: 666.
243. Shabsigh A, Korets R, Vora KC et al: Defining early morbidity of radical cystectomy for patients with bladder cancer using a standardized reporting methodology. *Eur Urol* 2009; **55**: 164.
244. Bochner BH, Dalbagni G, Sjoberg DD et al: Comparing open radical cystectomy and robot-assisted laparoscopic radical cystectomy: A randomized clinical trial. *Eur Urol* 2015; **67**: 1042.
245. Morgan TM, Keegan KA, Barocas DA et al: Predicting the probability of 90-day survival of elderly patients with bladder cancer treated with radical cystectomy. *J Urol* 2011; **186**: 829.
246. Hautmann RE, Gschwend JE, de Petriconi RC et al: Cystectomy for transitional cell carcinoma of the bladder: Results of a surgery only series in the neobladder era. *J Urol* 2006; **176**: 486.
247. Madersbacher S, Hochreiter W, Burkhard F et al: Radical cystectomy for bladder cancer today--a homogeneous series without neoadjuvant therapy. *J Clin Oncol* 2003; **21**: 690.
248. Shariat SF, Karakiewicz PI, Palapattu GS et al: Outcomes of radical cystectomy for transitional cell carcinoma of the bladder: A contemporary series from the bladder cancer research consortium. *J Urol* 2006; **176**: 2414.
249. Shariat SF, Palapattu GS, Amiel GE et al: Characteristics and outcomes of patients with carcinoma in situ only at radical cystectomy. *Urology* 2006; **68**: 538.
250. Weiss C, Wolze C, Engehausen DG et al: Radiochemotherapy after transurethral resection for high-risk t1 bladder cancer: An alternative to intravesical therapy or early cystectomy? *J Clin Oncol* 2006; **24**: 2318.
251. Kamat AM, Hegarty PK, Gee JR et al: IcuD-eau international consultation on bladder cancer 2012: Screening, diagnosis, and molecular markers. *European Urology* 2013; **63**: 4.
252. Stenzl A, Burger M, Fradet Y et al: Hexaminolevulinate guided fluorescence cystoscopy reduces recurrence in patients with nonmuscle invasive bladder cancer. *J Urol* 2010; **184**: 1907.
253. Burger M, Grossman HB, Droller M et al: Photodynamic diagnosis of non-muscle-invasive bladder cancer with hexaminolevulinate cystoscopy: A meta-analysis of detection and recurrence based on raw data. *Eur Urol* 2013; **64**: 846.
254. Heer R, Lewis R, Vadiveloo T et al: A randomized trial of photodynamic surgery in non-muscle-invasive bladder cancer. *NEJM Evidence* 2022; **1**: EVIDoA2200092.
255. Maisch P, Koziarz A, Vajgrt J et al: Blue versus white light for transurethral resection of non-muscle invasive bladder cancer. *Cochrane Database Syst Rev* 2021; **12**: Cd013776.
256. Enhanced visualization methods for first transurethral resection of bladder tumour in suspected non-muscle-invasive bladder cancer: A health technology assessment. *Ont Health Technol Assess Ser* 2021; **21**: 1.
257. Sari Motlagh R, Mori K, Laukhtina E et al: Impact of enhanced optical techniques at time of transurethral resection of bladder tumour, with or without single immediate intravesical chemotherapy, on recurrence rate of non-muscle-invasive bladder cancer: A systematic review and network meta-analysis of randomized trials. *BJU Int* 2021; **128**: 280.
258. Veeratterapillay R, Gravestock P, Nambiar A et al: Time to turn on the blue lights: A systematic review and meta-analysis of photodynamic diagnosis for bladder cancer. *Eur Urol Open Sci* 2021; **31**: 17.
259. Zhao H, Peng P, Luo Z et al: Comparison of hexaminolevulinate (hal) -guided versus white light transurethral resection for nmibc: A systematic review and meta-analysis of randomized controlled trials. *Photodiagnosis and Photodynamic Therapy* 2023; **41**: 103220.

260. Naselli A, Intorini C, Timossi L et al: A randomized prospective trial to assess the impact of transurethral resection in narrow band imaging modality on non-muscle-invasive bladder cancer recurrence. *Eur Urol* 2012; **61**: 908.
261. Geavlete B, Multescu R, Georgescu D et al: Narrow band imaging cystoscopy and bipolar plasma vaporization for large nonmuscle-invasive bladder tumors--results of a prospective, randomized comparison to the standard approach. *Urology* 2012; **79**: 846.
262. Tschirdewahn S, Harke NN, Hirner L et al: Narrow-band imaging assisted cystoscopy in the follow-up of patients with transitional cell carcinoma of the bladder: A randomized study in comparison with white light cystoscopy. *World J Urol* 2020; **38**: 1509.
263. Murakami Y, Matsumoto K, Miyake M et al: Real-world treatment patterns and oncological outcomes in early relapse and refractory disease after bacillus calmette-guérin failure in non-muscle-invasive bladder cancer. *Int J Urol* 2022; **29**: 1195.
264. Lai LY, Tafuri SM, Ginier EC et al: Narrow band imaging versus white light cystoscopy alone for transurethral resection of non-muscle invasive bladder cancer. *Cochrane Database of Systematic Reviews* 2022.
265. Howard JM, Cook GS, Tverye A et al: Outcomes of patients with bacillus calmette-guérin (bcg)-unresponsive non-muscle invasive bladder cancer as defined by the u.S. Food and drug administration. *Bladder Cancer* 2022; **8**: 303.
266. Gravestock P, Coulthard N, Veeratterapillay R and Heer R: Systematic review and meta-analysis of narrow band imaging for non-muscle-invasive bladder cancer. *Int J Urol* 2021; **28**: 1212.
267. Olsen LH and Genster HG: Prolonging follow-up intervals for non-invasive bladder tumors: A randomized controlled trial. *Scand J Urol Nephrol Suppl* 1995; **172**: 33.
268. Matsumoto K, Kikuchi E, Horiguchi Y et al: Late recurrence and progression in non-muscle-invasive bladder cancers after 5-year tumor-free periods. *Urology* 2010; **75**: 1385.
269. Holmäng S and Ströck V: Should follow-up cystoscopy in bacillus calmette-guérin-treated patients continue after five tumour-free years? *Eur Urol* 2012; **61**: 503.
270. Sternberg IA, Keren Paz GE, Chen LY et al: Upper tract imaging surveillance is not effective in diagnosing upper tract recurrence in patients followed for nonmuscle invasive bladder cancer. *J Urol* 2013; **190**: 1187.
271. Park DS, Hwang JH, Gong IH et al: An analysis of the efficacy, safety, and cost-effectiveness of fulguration under local anesthesia for small-sized recurrent masses: A comparative analysis to transurethral resection of bladder tumors in a matched cohort. *J Endourol* 2013; **27**: 1240.
272. Herr HW: Outpatient flexible cystoscopy and fulguration of recurrent superficial bladder tumors. *J Urol* 1990; **144**: 1365.
273. Klein FA and Whitmore WF, Jr.: Bladder papilloma: Therapeutic and cost effect of outpatient department management. *Urology* 1981; **18**: 247.
274. Rose KM, Huelster HL, Meeks JJ et al: Circulating and urinary tumour DNA in urothelial carcinoma - upper tract, lower tract and metastatic disease. *Nat Rev Urol* 2023; **20**: 406.
275. Carosella ED, Ploussard G, LeMaoult J and Desgrandchamps F: A systematic review of immunotherapy in urologic cancer: Evolving roles for targeting of ctla-4, pd-1/pd-l1, and hla-g. *Eur Urol* 2015; **68**: 267.
276. Shore ND, Boorjian SA, Canter DJ et al: Intravesical rad-ifn α /syn3 for patients with high-grade, bacillus calmette-guerin-refractory or relapsed non-muscle-invasive bladder cancer: A phase ii randomized study. *J Clin Oncol* 2017; **35**: 3410.
277. Chamie K, Chang SS, Kramolowsky E et al: IL-15 superagonist NAI in BCG-unresponsive non-muscle-invasive bladder cancer. *NEJM Evid* 2023; **2**.
278. Del Giudice F, Barchetti G, De Berardinis E et al: Prospective assessment of vesical imaging reporting and data system (vi-rads) and its clinical impact on the management of high-risk non-muscle-invasive bladder cancer patients candidate for repeated transurethral resection. *Eur Urol* 2020; **77**: 101.



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